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Microscopic Simulation of Solid State Sintering Regarding Irregularly Shaped Powder Particles

Defense of Doctoral Thesis

Max Weiner

Institute of Metal Forming, TU Bergakademie Freiberg

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Neck Size

Introduction

Definition of Sintering

Sintering

- Is a heat treatment process to alter microstructure of powder compacts for achieving certain properties (mainly mechanical strength).
- Is a thermodynamically voluntary process driven by reduction of energy stored in defects (especially surfaces and grain boundaries).

The system is in non-equilibrium thermodynamic state and evolves towards equilibrium (seldom ever reaching it).

Equilibrium vs. Non-Equilibrium Thermodynamics

Equilibrium

- The equilibrium state of a system is characterized by a minimum of the corresponding thermodynamic potential (or a maximum of overall entropy).
- A system outside equilibrium will voluntarily change its state approaching the equilibrium.

Problem Entropy and temperature are only well-defined in equilibrium.

Attempts

- Assuming a local equilibrium state (or accompanying equilibrium state).
- Just assuming the validity of those definitions outside equilibrium.

Internal Variables of State

Variables of State

- Natural variables of the thermodynamic potentials: T or S , p or V , n or μ .
- Define the macroscopic state of the system.
- Measurable and controllable.

Internal Variables of State

- Arbitrary variables characterizing microscopic state at arbitrary level of detail.
- Account for *our lack of knowledge* about microstructural state of the system.
- May be measurable, but usually out of direct control.

Encouragement

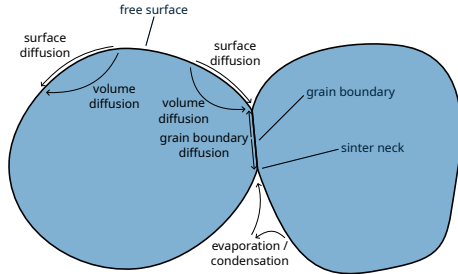
Perhaps, after all, the wise man's attitude towards thermodynamics should be to have nothing to do with it. To deal with thermodynamics is to look for trouble.
— Maugin (1999)

State of the Art

Sintering Models

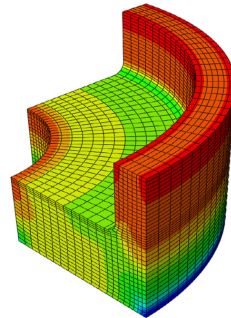
Microscopic Models

- Regard distinct particles.
- Model diffusional fluxes and evolution of particle shape.

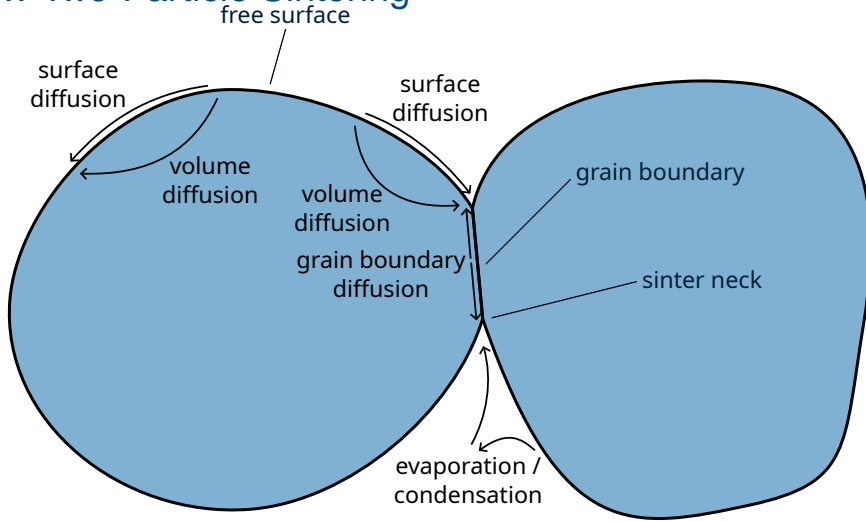


Macroscopic Models

- Regard components consisting of "continuous" material.
- Model densification, stresses and strains.



Overview: Two-Particle Sintering



Overview on Sintering Modelling Approaches

Direct Representation

- discretization of interface line
- points move with evolving interface
- interface tracked by point position
- discretization depends on shape requirements

Indirect Representation

- discretization of solution space
- points fixed (except remeshing)
- interface tracked by state function value
- discretization depends on gradient requirements

Sharp Interface

- interface is sharp and unsteady jump
- interface energy is localized

Diffuse Interface

- interface is smooth and steady transition
- interface energy is blurred

Overview on Sintering Modelling Approaches

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Indirect Representation

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Sharp Interface

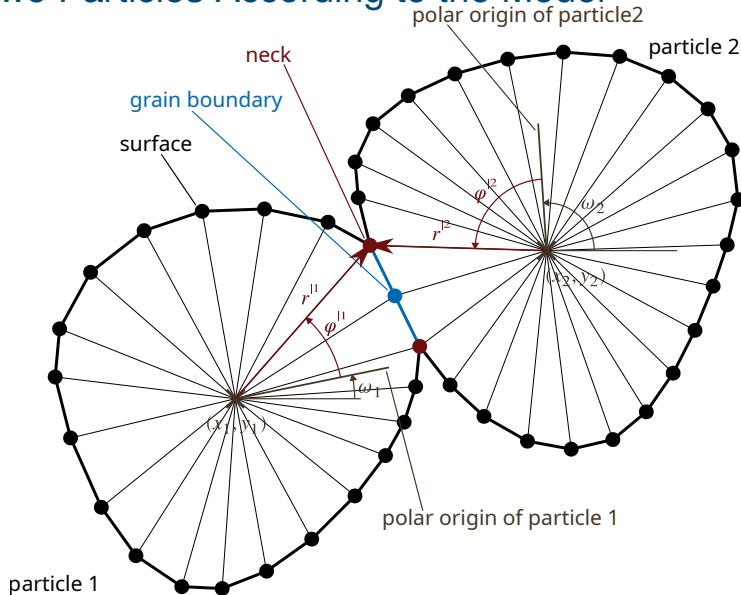
- interface is sharp and unsteady jump
- interface energy is localized

Diffuse Interface

- interface is smooth and steady transition
- interface energy is blurred

Sintering Model Development

Overview: Two Particles According to the Model



Thermodynamic Extremal Principle

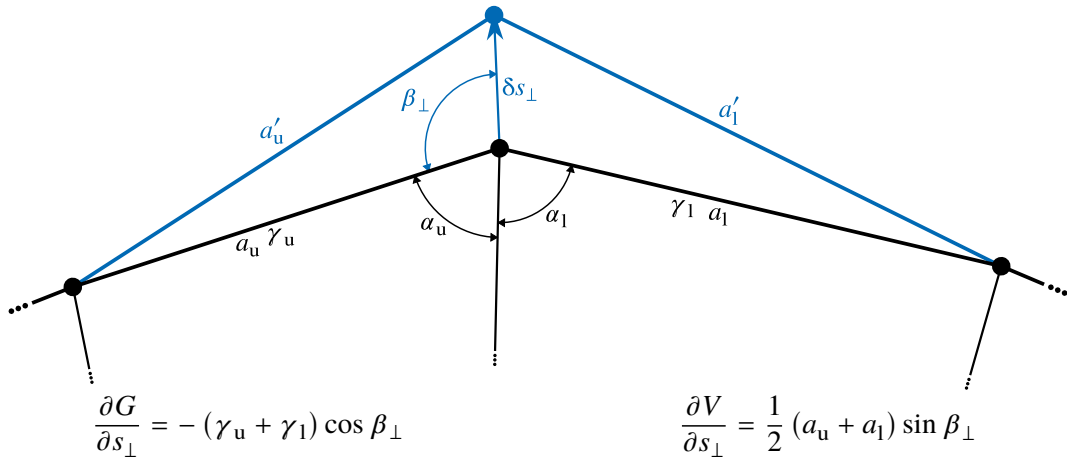
Thermodynamic Extremal Principle

System evolution will follow the path of maximum dissipation with respect to all constraints.

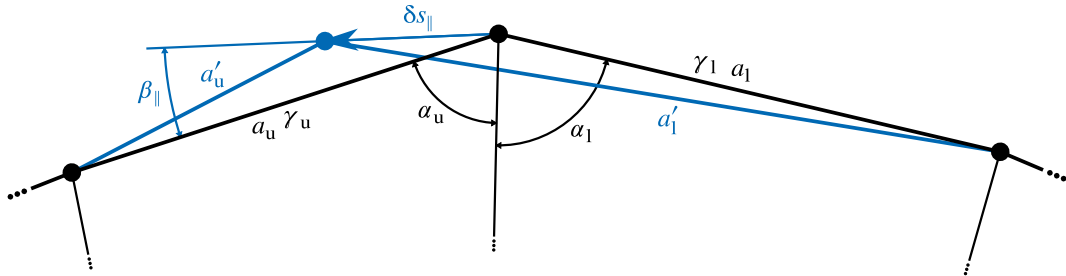
Thermodynamic Dissipation	$\mathcal{D}(X, \mathbf{x}, \dot{\mathbf{x}}) = -\frac{\partial G(X, \mathbf{x}, \xi)}{\partial \mathbf{x}} \cdot \dot{\mathbf{x}} \rightarrow \max_{\dot{\mathbf{x}}}$
Kinetic Dissipation	$\mathcal{D}(X, \mathbf{x}, \dot{\mathbf{x}}) - \mathcal{Q}(X, \mathbf{x}, \mathbf{j}) = 0$
Constraints	$\mathbf{g}(X, \mathbf{x}, \dot{\mathbf{x}}, \mathbf{j}, \xi) = 0$ (volume balance, contact conditions, ...)

Onsager 1931; Ziegler 1963; Svoboda et al. 1991; Fischer et al. 2014; Hackl et al. 2020, ...

Normal Node Shifting



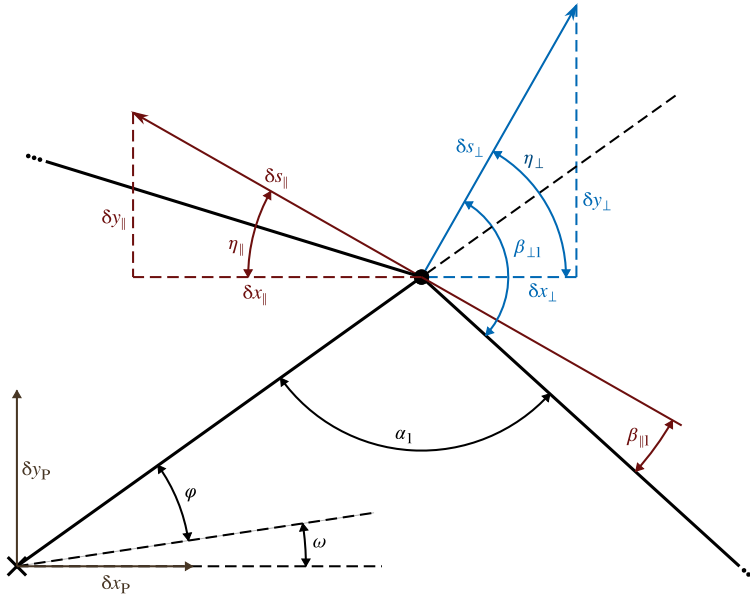
Tangential Node Shifting



$$\frac{\partial G}{\partial s_{\parallel}} = \lim_{\delta s_{\parallel} \rightarrow 0} \frac{\delta G_{\parallel}}{\delta s_{\parallel}} = -(\gamma_u - \gamma_1) \cos \beta_{\parallel}$$

$$\frac{\partial V}{\partial s_{\parallel}} = \lim_{\delta s_{\parallel} \rightarrow 0} \frac{\delta V_{\parallel u} - \delta V_{\parallel 1}}{\delta s_{\parallel}} = \frac{1}{2} (a_u - a_1) \sin \beta_{\parallel}$$

Contact Conditions: Absolute Node Shift

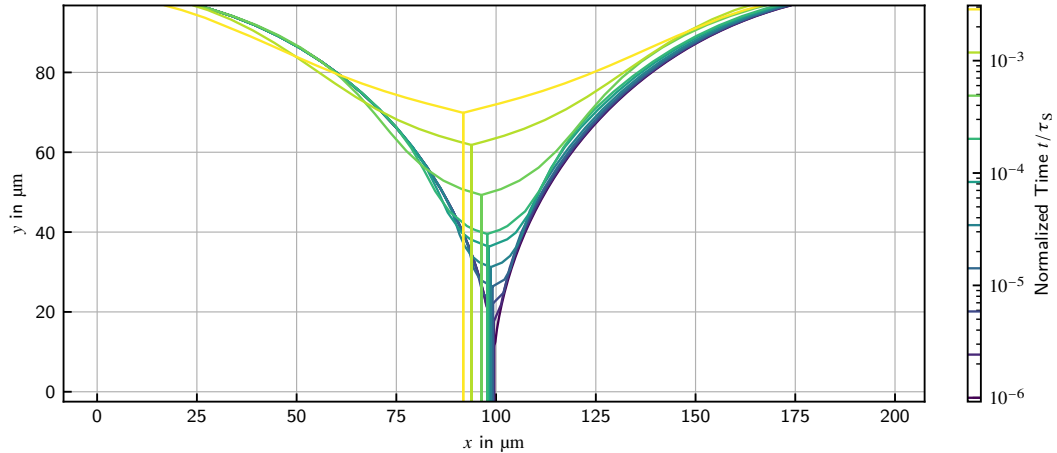


Two-Particle Contacts

Two-Particle Contacts

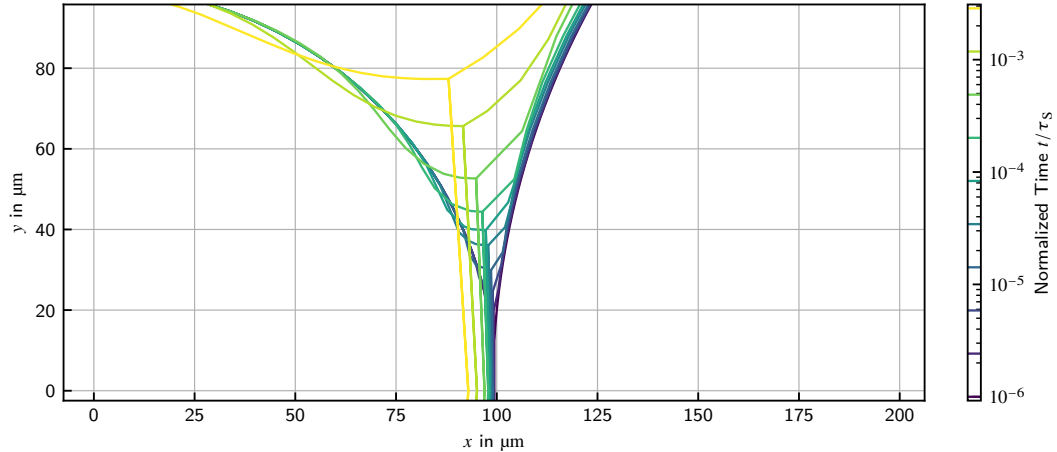
Particle-Size-Ratio

Geometry Evolution – Particle-Size-Ratio r_2/r_1



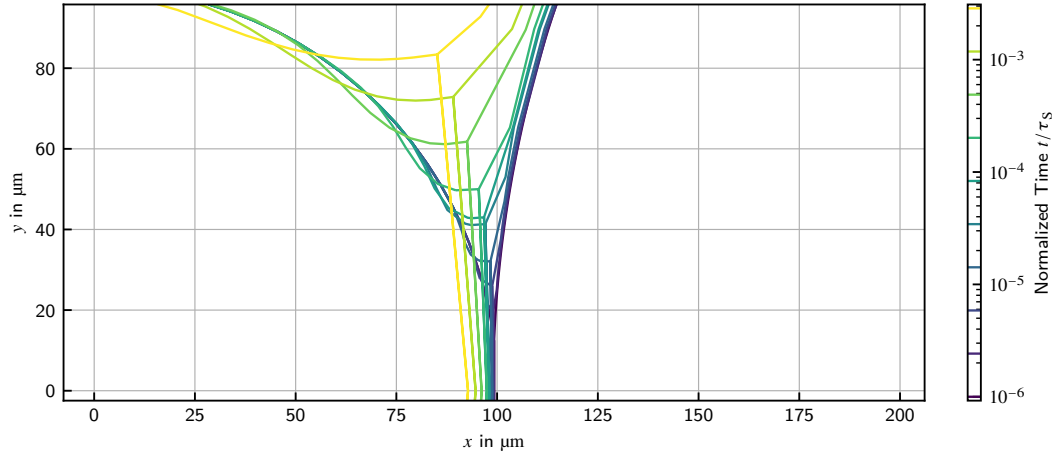
$$r_2/r_1 = 1$$

Geometry Evolution – Particle-Size-Ratio r_2/r_1



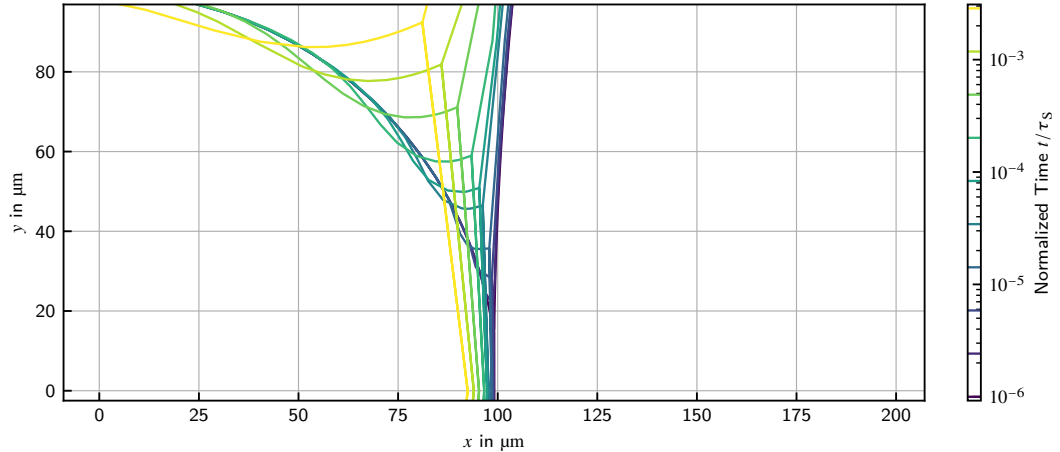
$$r_2/r_1 = 2$$

Geometry Evolution – Particle-Size-Ratio r_2/r_1

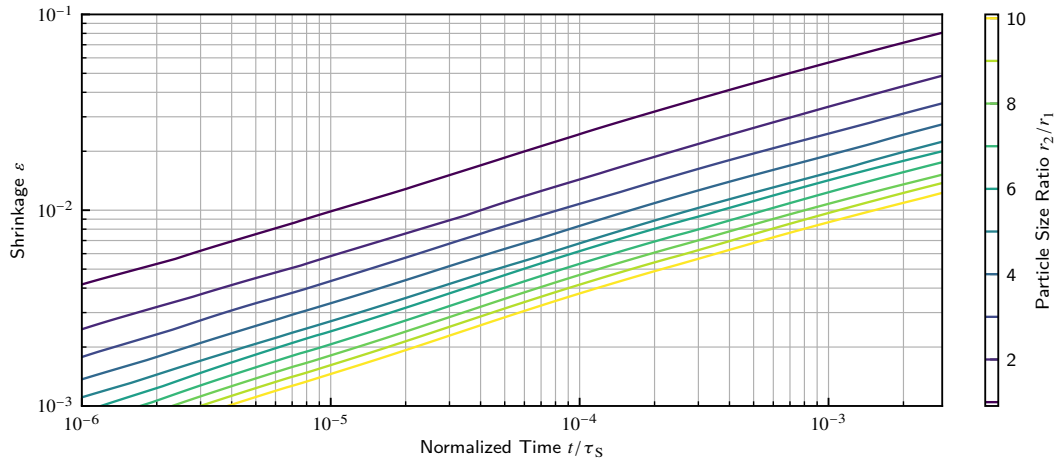


$$r_2/r_1 = 3$$

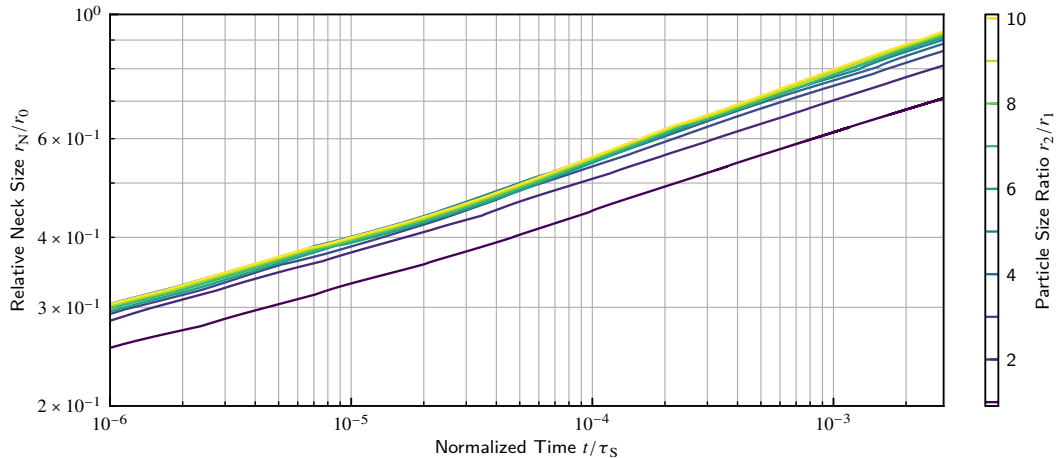
Geometry Evolution – Particle-Size-Ratio r_2/r_1



Shrinkage Evolution – Particle-Size-Ratio r_2/r_1

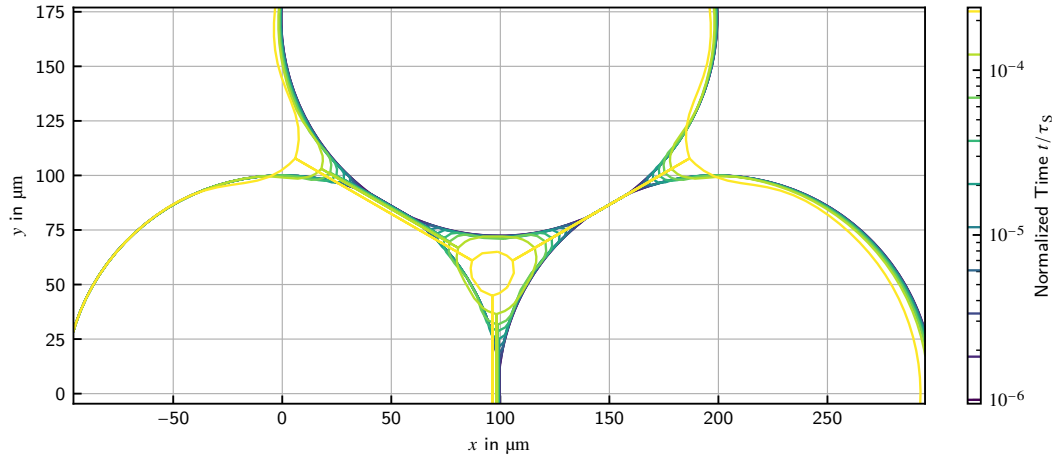


Neck Size Evolution – Particle-Size-Ratio r_2/r_1

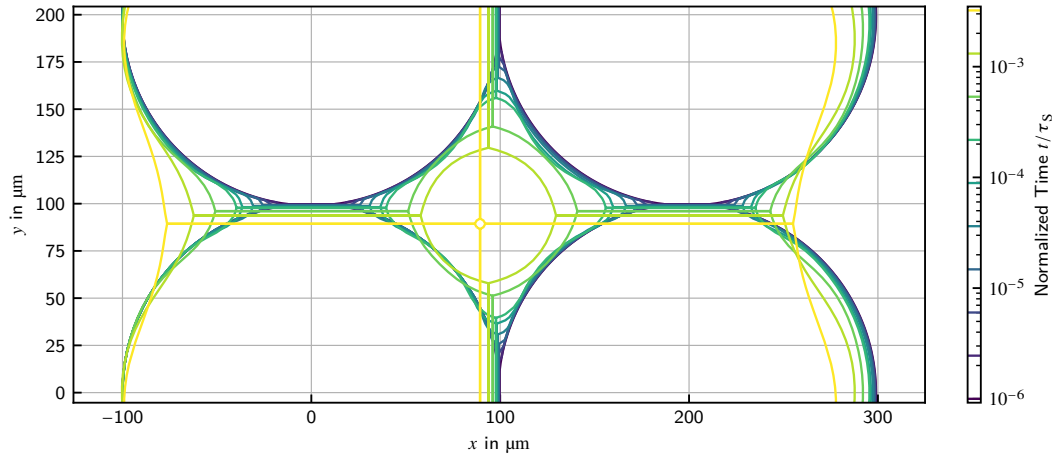


Multi-Particle Contacts

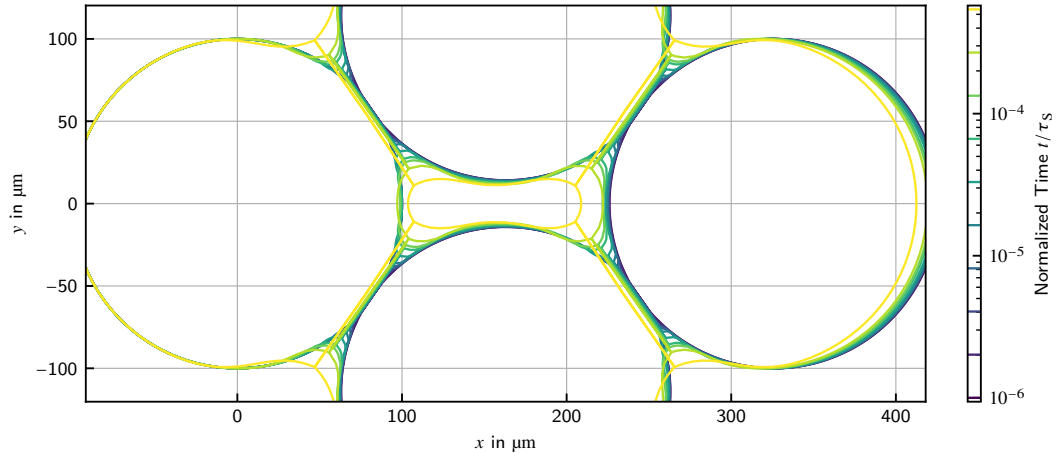
Geometry Evolution - Triangle Packing



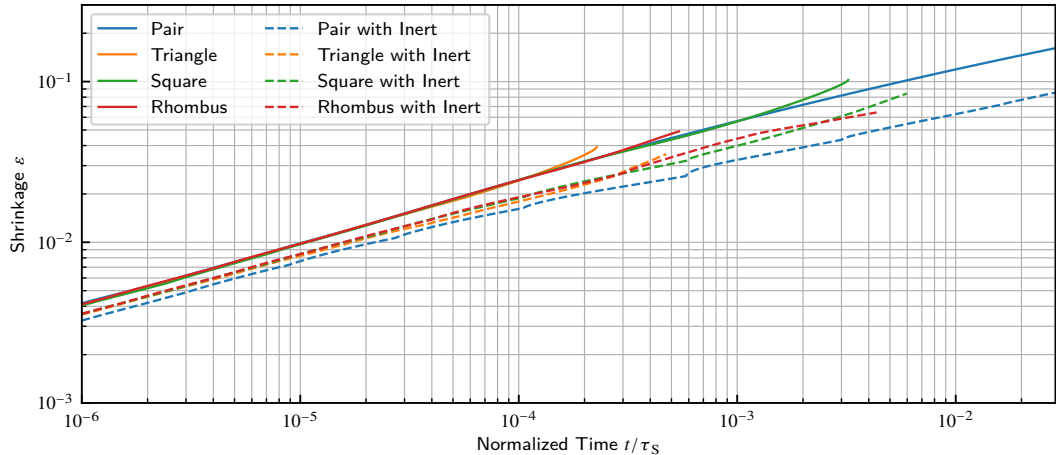
Geometry Evolution - Square Packing



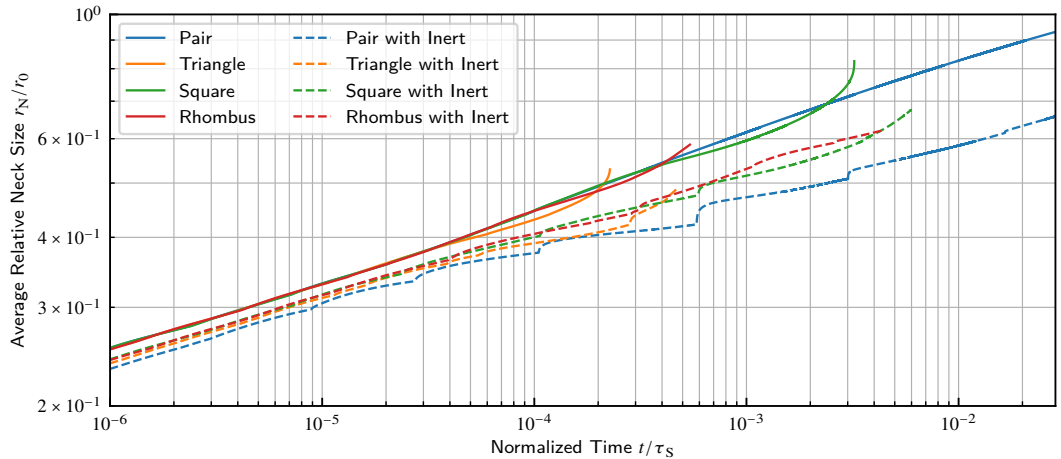
Geometry Evolution - Rhombus Packing



Shrinkage



Average Neck Size



Statistical Powder Modeling

Particle Description Using Shape Function

Base Function

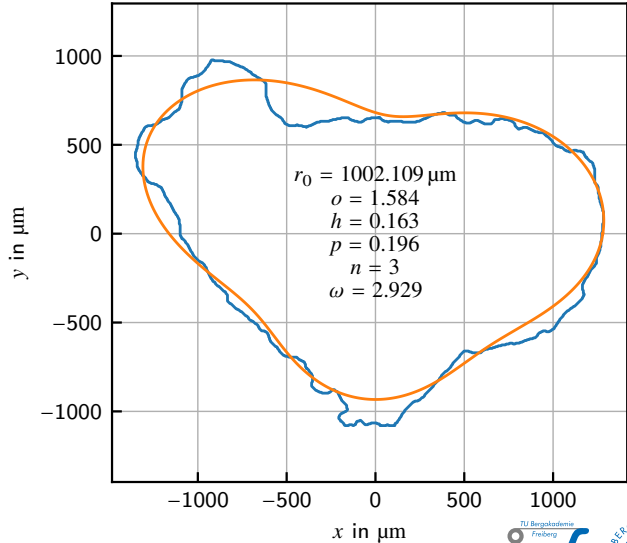
$$r(\varphi) = r_0 \cdot \left[\prod_i f_i(\varphi) \right]$$

Ovality Shape Function

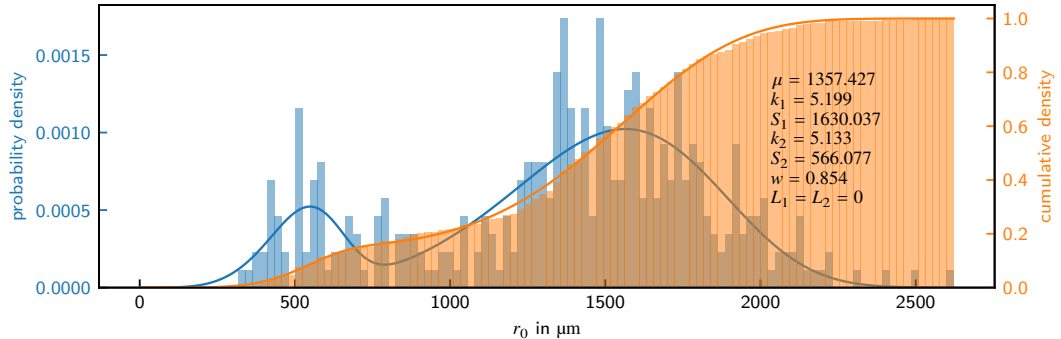
$$f_o = \sqrt{\frac{o}{o^2 \sin^2 \varphi + \cos^2 \varphi}}$$

First-Order Wave Function

$$f_1 = h_1 \cdot \cos(n_1 \varphi + 2\pi p_1) + 1$$

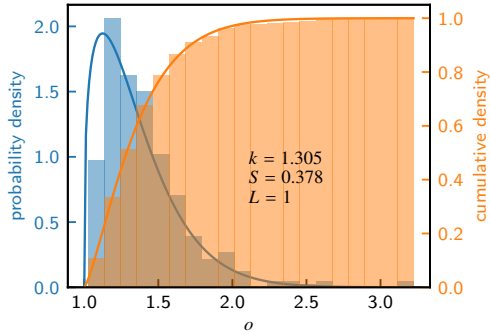


Statistical Results on Particle Morphology

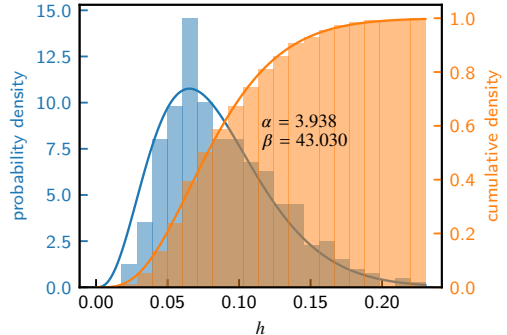


Particle Size
Bimodal Mixed Weibull-Distribution

Statistical Results on Particle Morphology

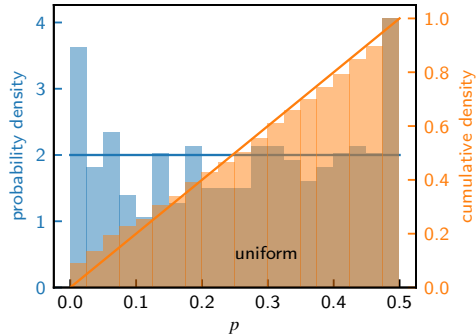


Ovality
Weibull-Distribution

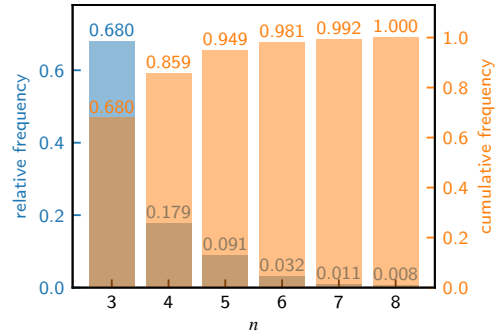


Wave Height
Beta-Distribution

Statistical Results on Particle Morphology



Phase Shift
Uniform Distribution

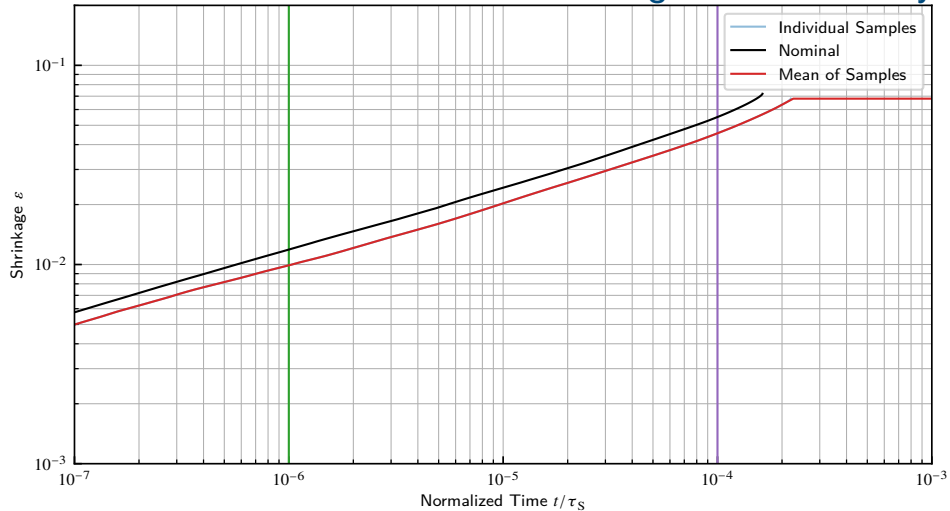


Wave Count
Categorical Distribution

Randomized Simulations – Monte-Carlo Approach

- Random sampling of radius, ovality and shape parameters for 3 distinct particles.
- Generation of initial contact state.
- Simulation till end time reached or pore closed.
- Repeating several hundred times and collecting results.
- Easy parallelization due to independent simulations.

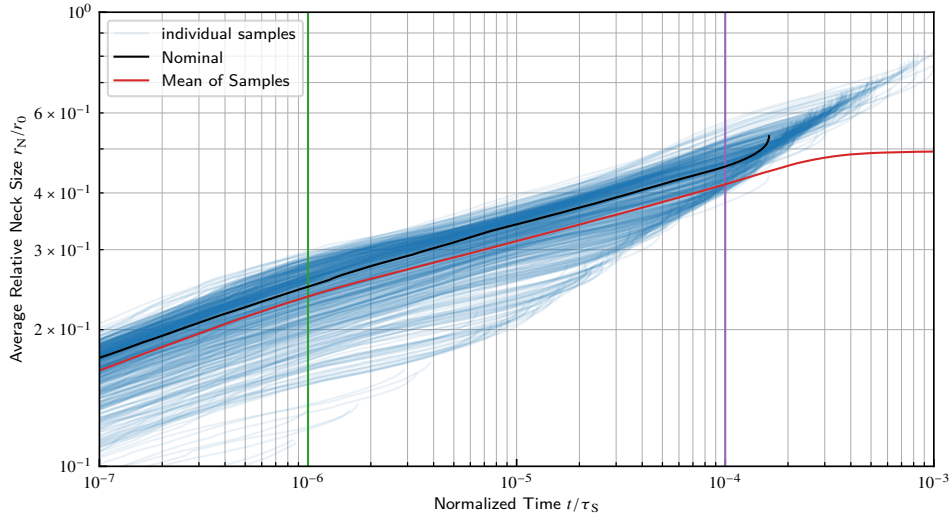
Randomized Simulation Results – Building a Curve Family Plot



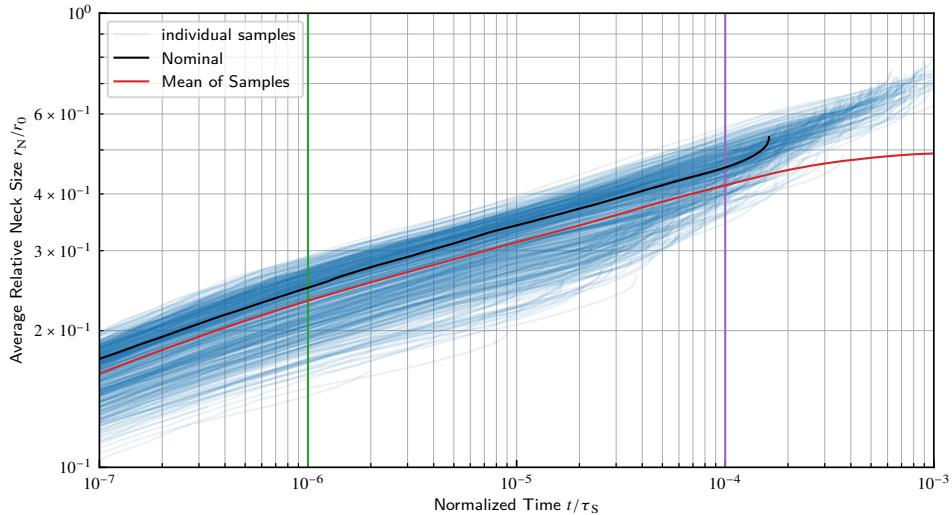
Statistical Powder Modeling

Shrinkage

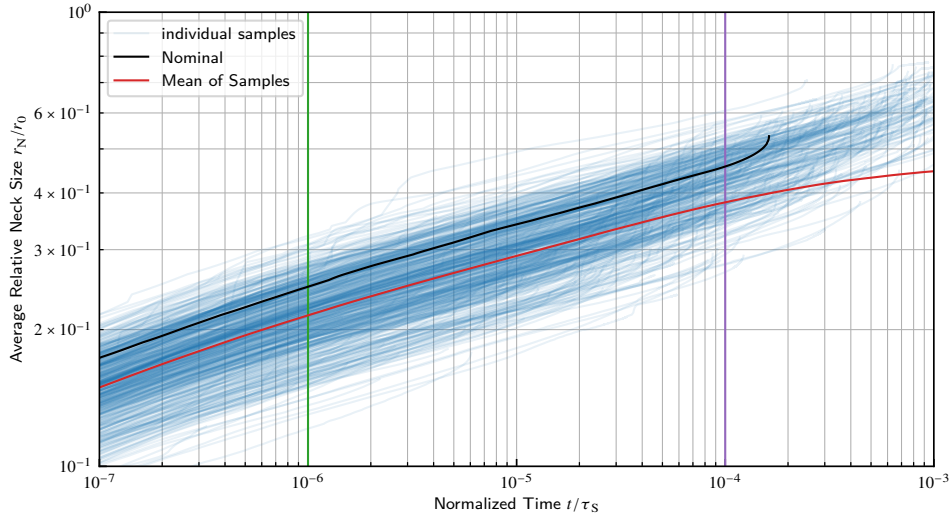
Randomized Simulation Results – Neck Size – Circular Particles



Randomized Simulation Results – Neck Size – Oval Particles



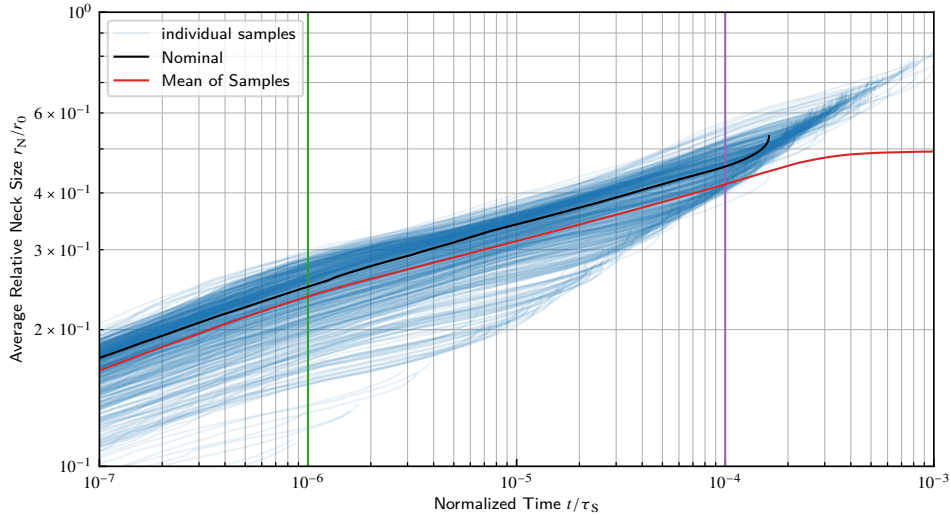
Randomized Simulation Results – Neck Size – Waved Particles



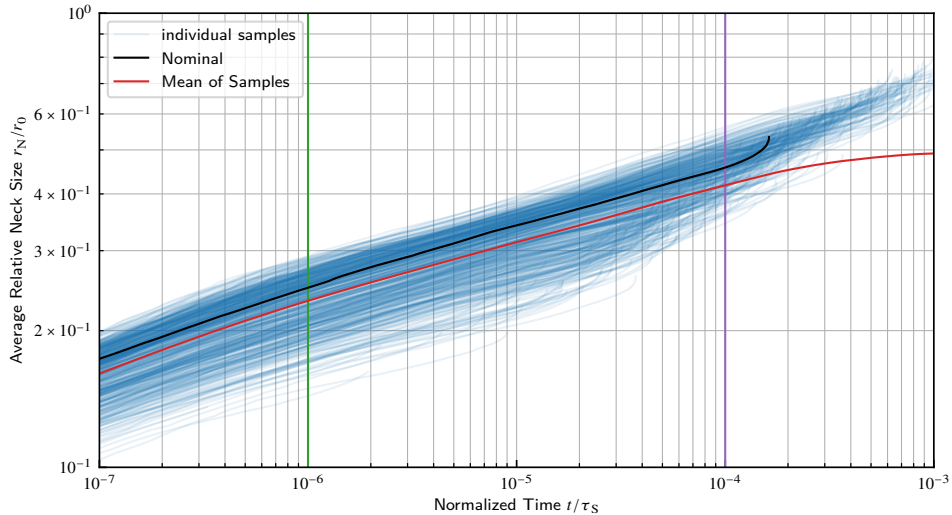
Statistical Powder Modeling

Neck Size

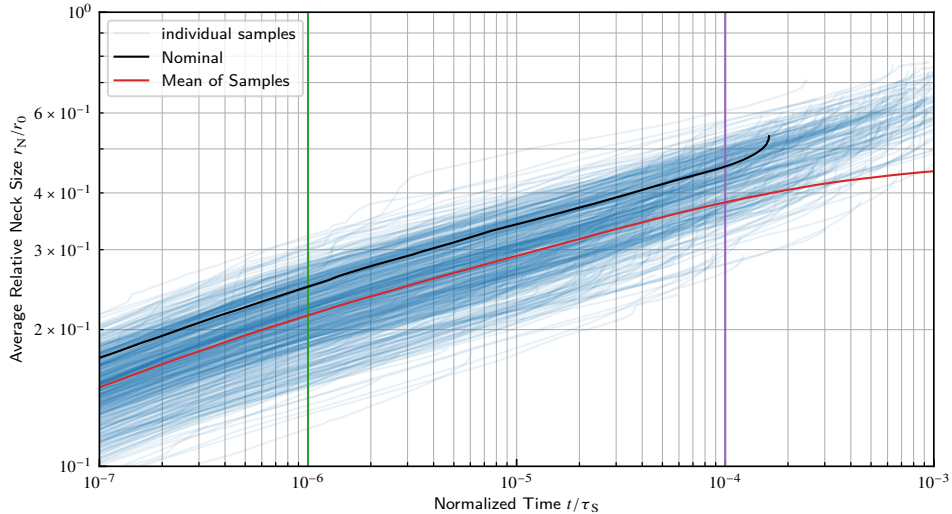
Randomized Simulation Results – Neck Size – Circular Particles



Randomized Simulation Results – Neck Size – Oval Particles



Randomized Simulation Results – Neck Size – Waved Particles





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Thank you for your attention!

Max Weiner
Institute of Metal Forming
TU Bergakademie Freiberg
Bernhard-von-Cotta-Str. 4
DE-09599 Freiberg
Phone: 049 3731 39 2952
Mail: max.weiner@imf.tu-freiberg.de



Appendix

Appendix Contents

7. Sintering Model Development

8. Model Implementation

9. Two-Particle Contacts

Surface-Boundary-Energy-Ratio

Surface-Boundary-Diffusion-Ratio

Particle-Size-Ratio

Surface-Energy-Asymmetry

Diffusion-Asymmetry

Ovality – Tip-Tip-Configuration

Ovality – Tip-Flank-Configuration

Ovality – Flank-Flank-Configuration

10. Multi-Particle Contacts

11. Statistical Powder Modeling

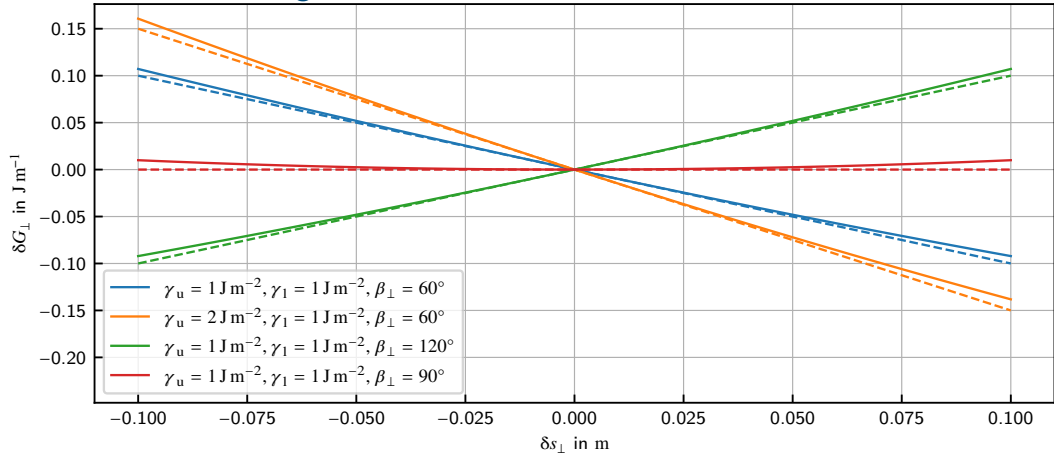
Shrinkage

Neck Size

Appendix

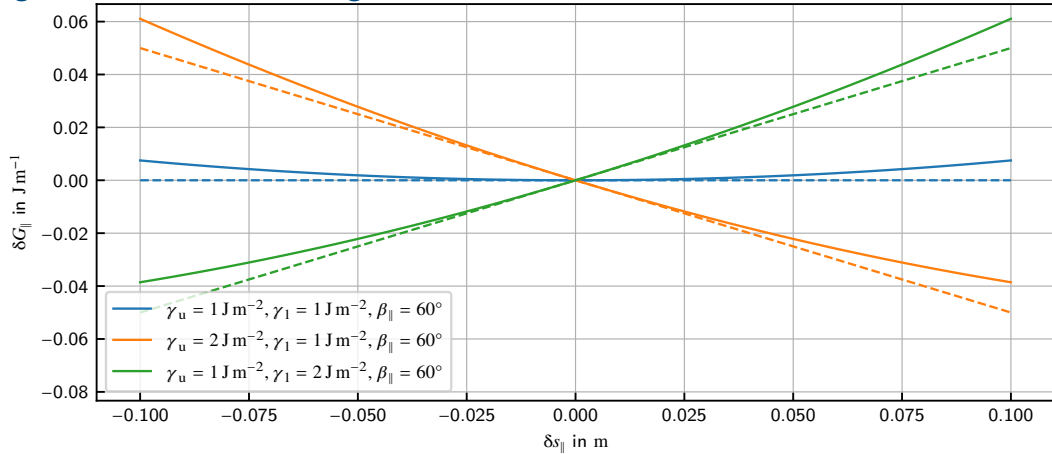
Sintering Model Development

Normal Node Shifting



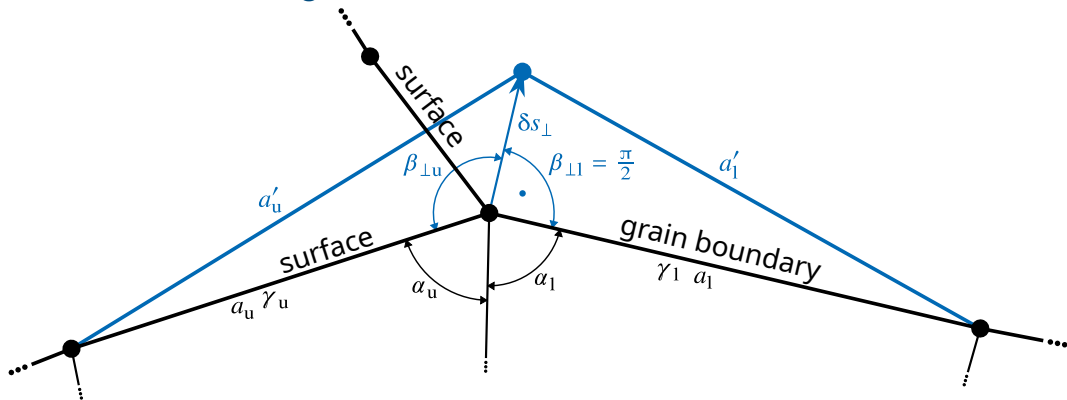
$$\frac{\partial G}{\partial s_{\perp}} = -(\gamma_u + \gamma_l) \cos \beta_{\perp}$$

Tangential Node Shifting – Potential



$$\frac{\partial G}{\partial s_{||}} = \lim_{\delta s_{||} \rightarrow 0} \frac{\delta G_{||}}{\delta s_{||}} = -(\gamma_u - \gamma_1) \cos \beta_{||}$$

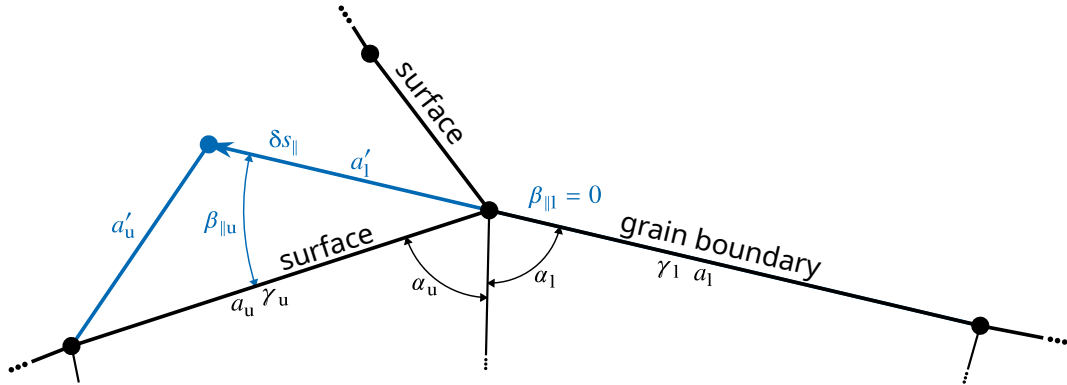
Normal Node Shifting – Neck



$$\frac{\partial G}{\partial s_{\perp}} = -(\gamma_u \cos \beta_{\perp u} - \gamma_l \cos \beta_{\perp l})$$

$$\frac{\partial V}{\partial s_{\perp}} = \frac{1}{2} (a_u \sin \beta_{\perp u} + a_l \sin \beta_{\perp l})$$

Tangential Node Shifting – Neck



$$\frac{\partial G}{\partial s_{\parallel}} = -(\gamma_u \cos \beta_{\parallel u} - \gamma_1 \cos \beta_{\parallel 1})$$

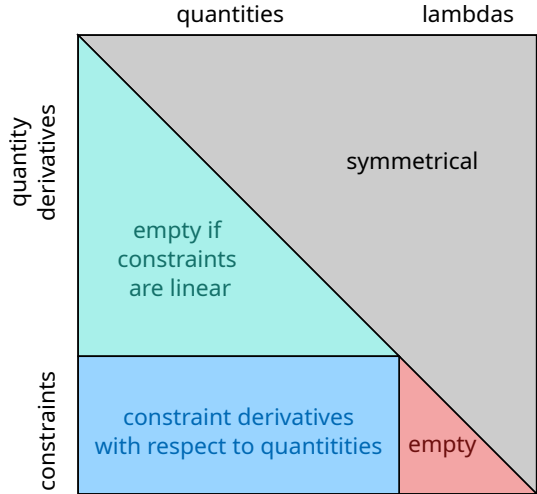
$$\frac{\partial V}{\partial s_{\parallel}} = \frac{1}{2} (a_u \sin \beta_{\parallel u} - a_1 \sin \beta_{\parallel 1})$$

Appendix

Model Implementation

Equation System Structure

- Non-linear equation system.
- Sparse symmetrical gradient matrix.
- Block structure regarding quantities (state variables, fluxes) and constraints (with their lambdas).



Appendix

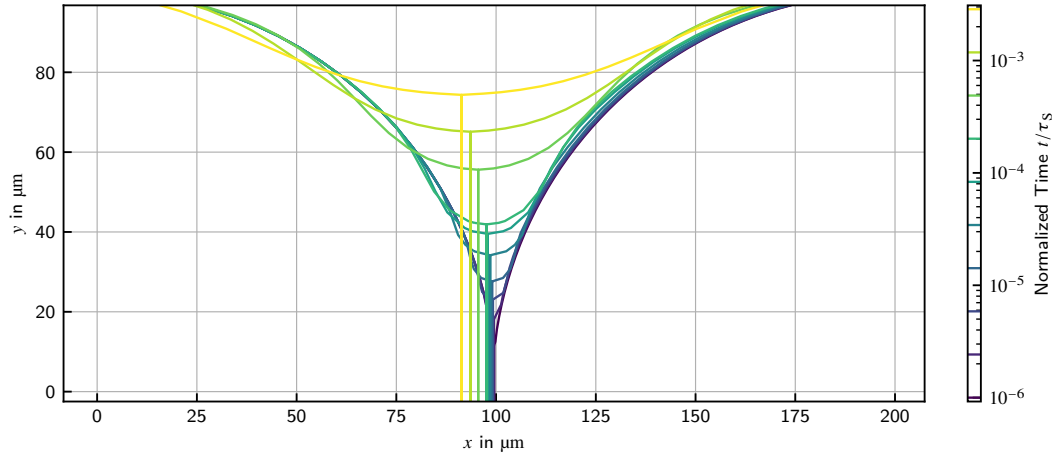
Two-Particle Contacts

Appendix

Two-Particle Contacts

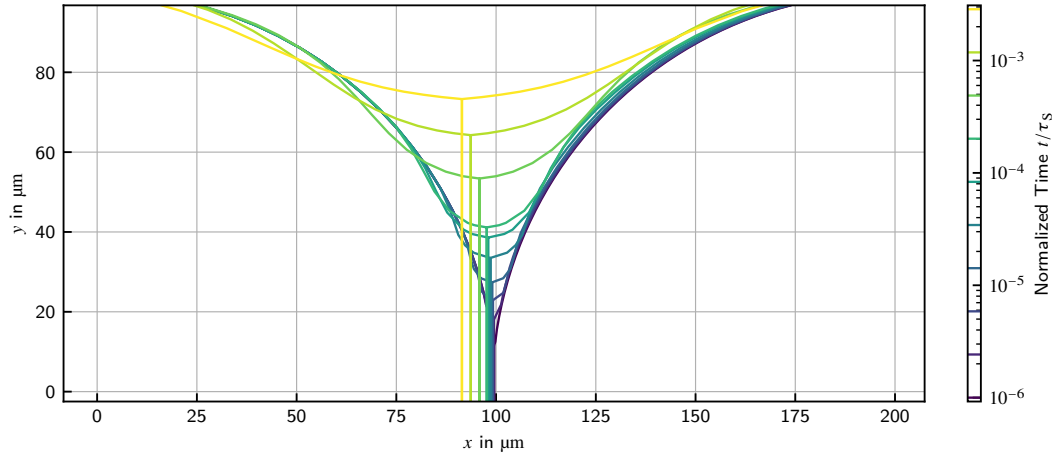
Surface-Boundary-Energy-Ratio

Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



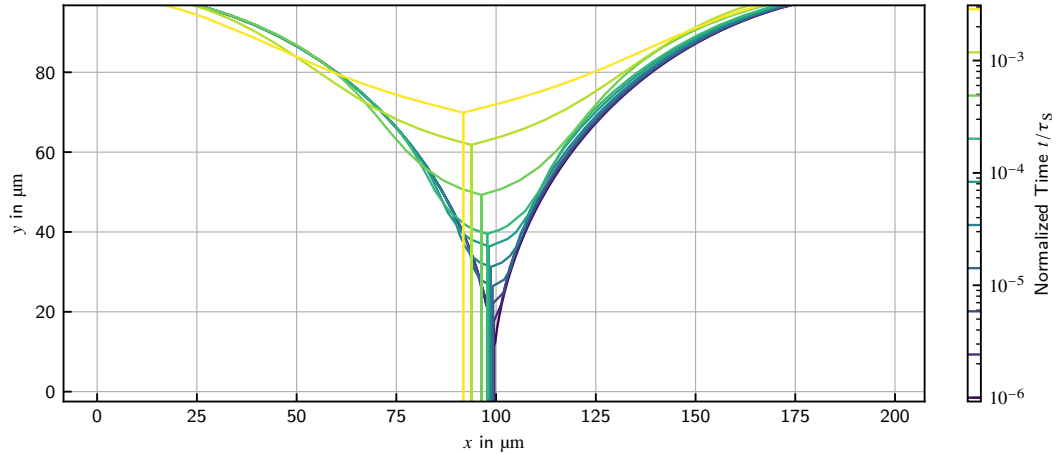
$$\gamma_{GB}/\gamma_S = 0.1$$

Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



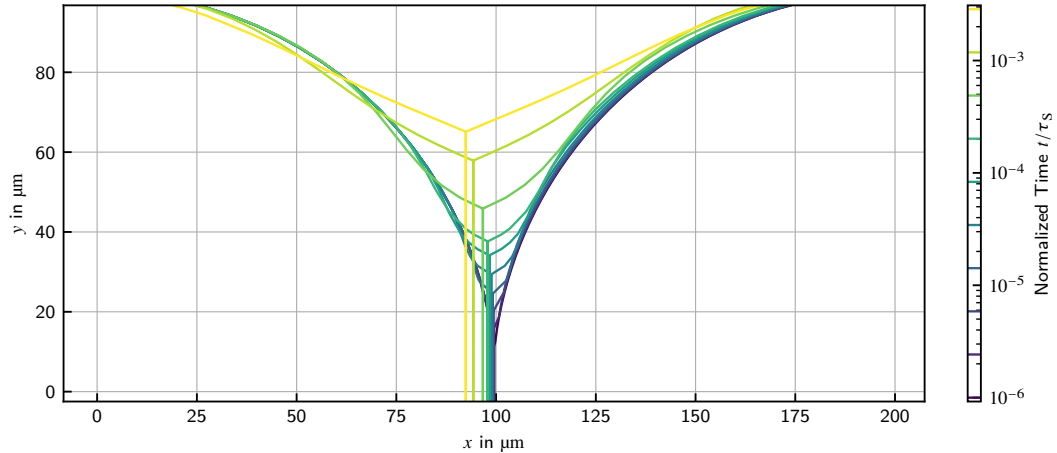
$$\gamma_{GB}/\gamma_S = 0.2$$

Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



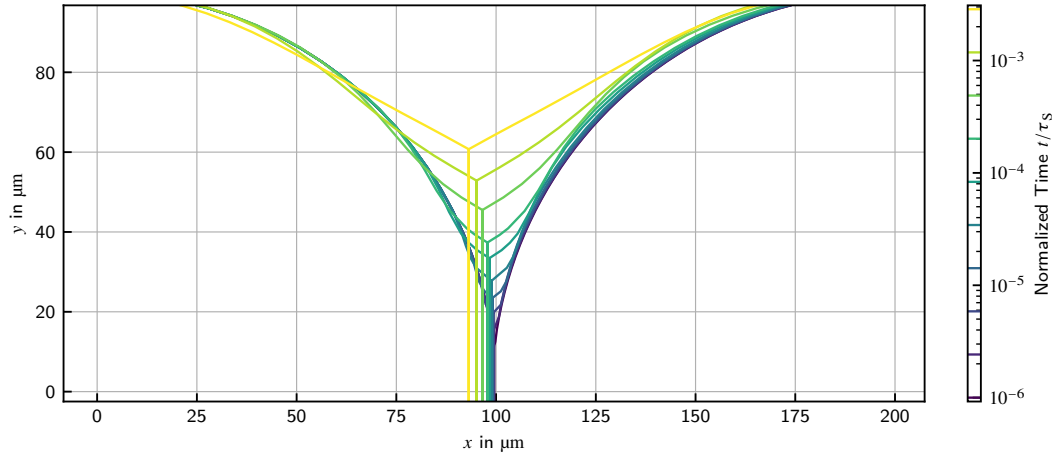
$$\gamma_{GB}/\gamma_S = 0.5$$

Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



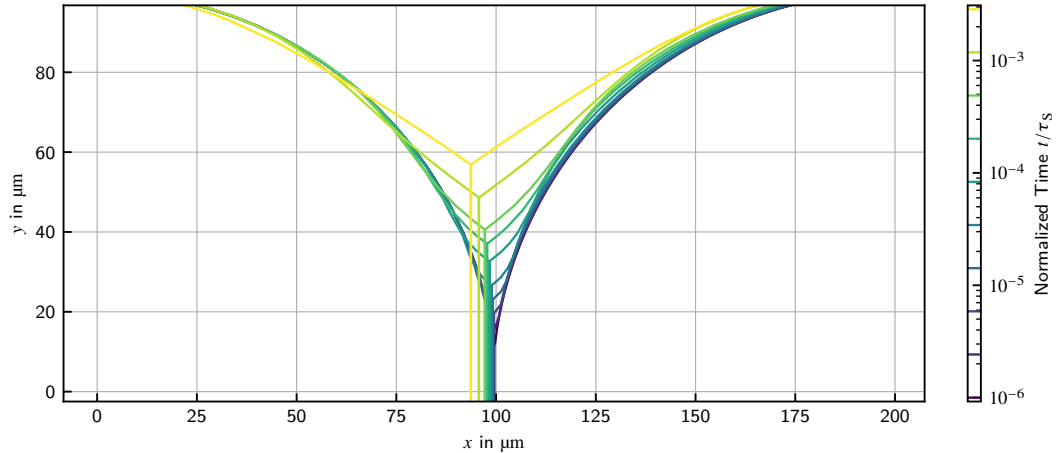
$$\gamma_{GB}/\gamma_S = 0.8$$

Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S

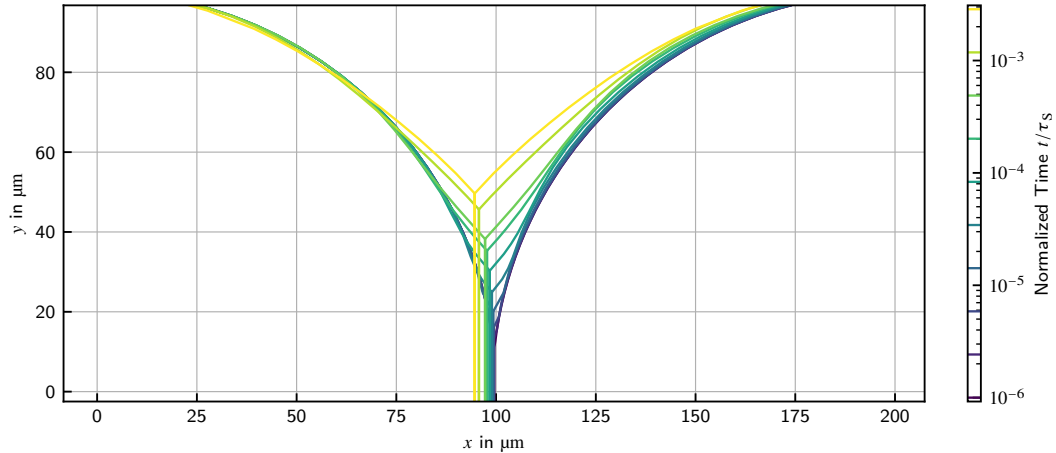


$$\gamma_{GB}/\gamma_S = 1.0$$

Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S

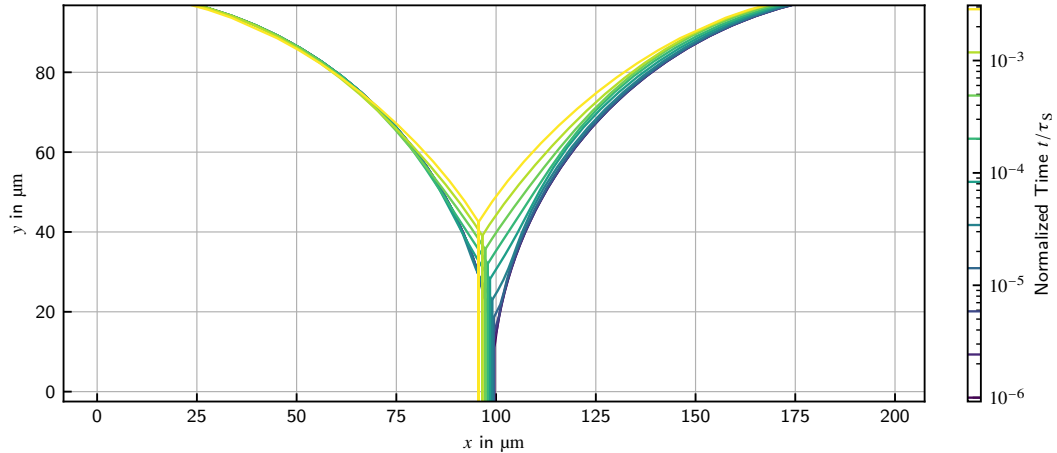


Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



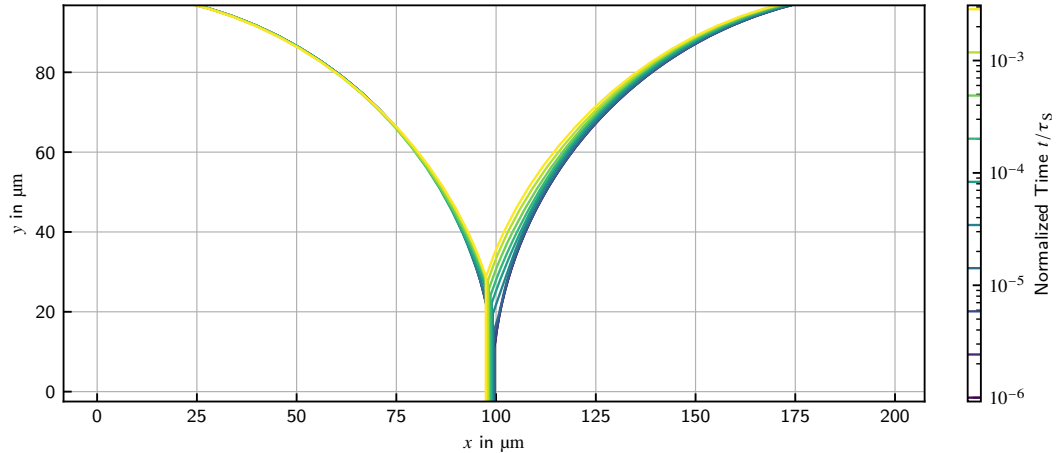
$$\gamma_{GB}/\gamma_S = 1.5$$

Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



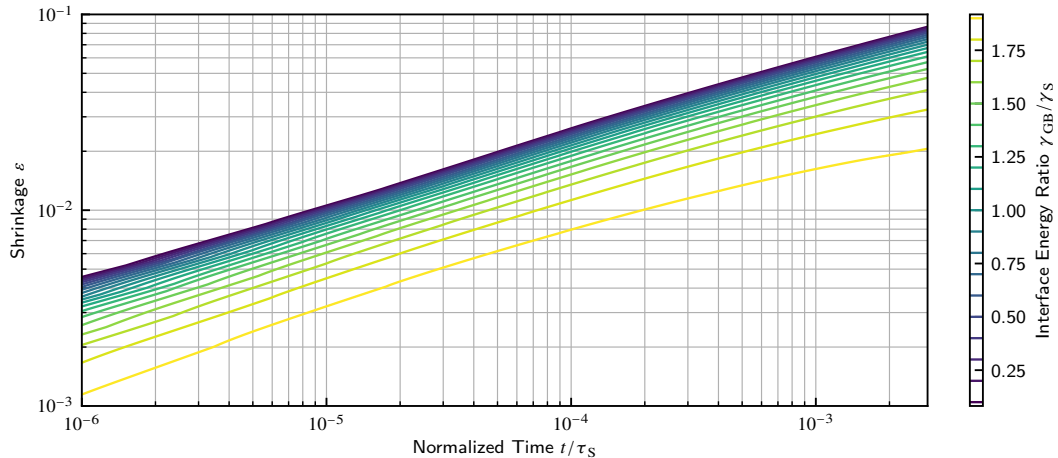
$$\gamma_{GB}/\gamma_S = 1.7$$

Geometry Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S

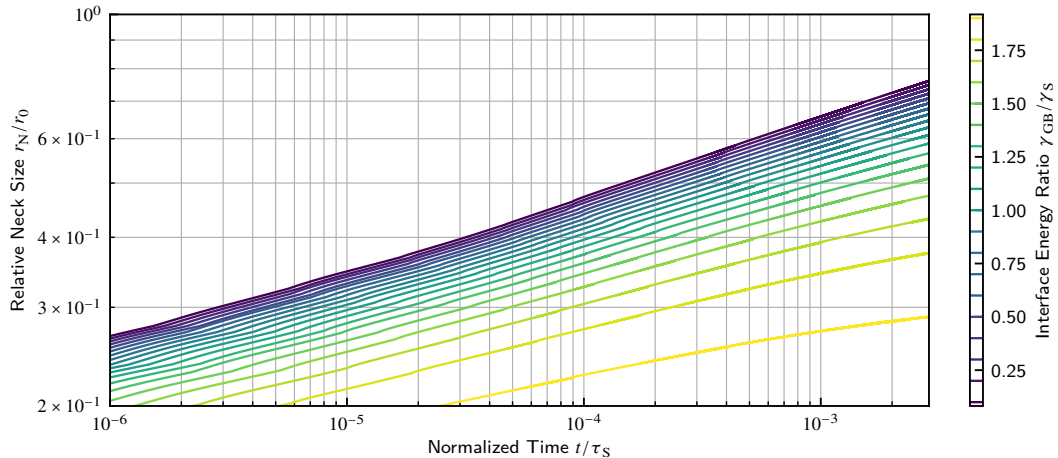


$$\gamma_{GB}/\gamma_S = 1.9$$

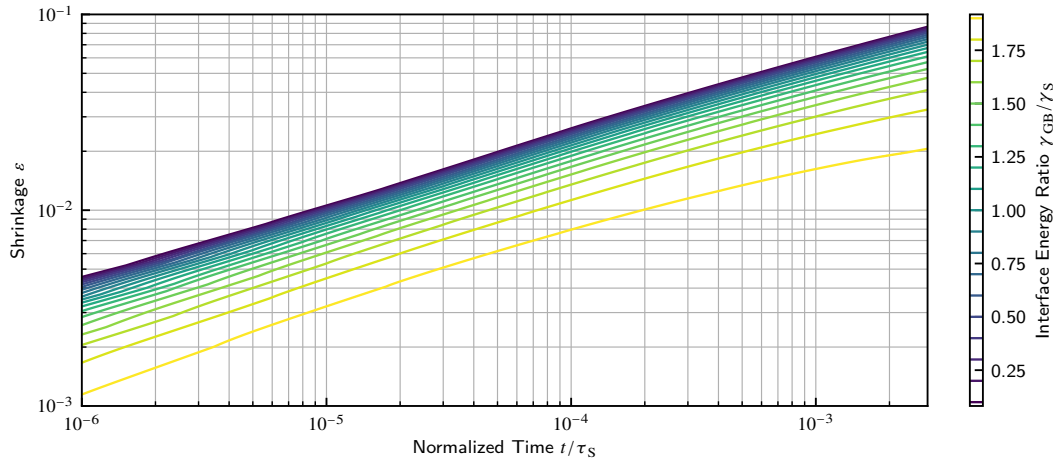
Shrinkage Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



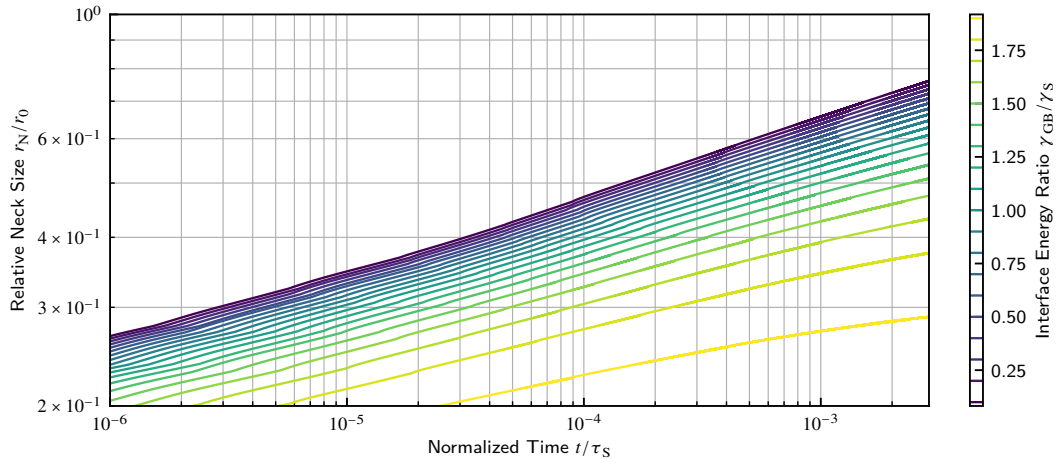
Neck Size Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



Shrinkage Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S



Neck Size Evolution – Surface-Boundary-Energy-Ratio γ_{GB}/γ_S

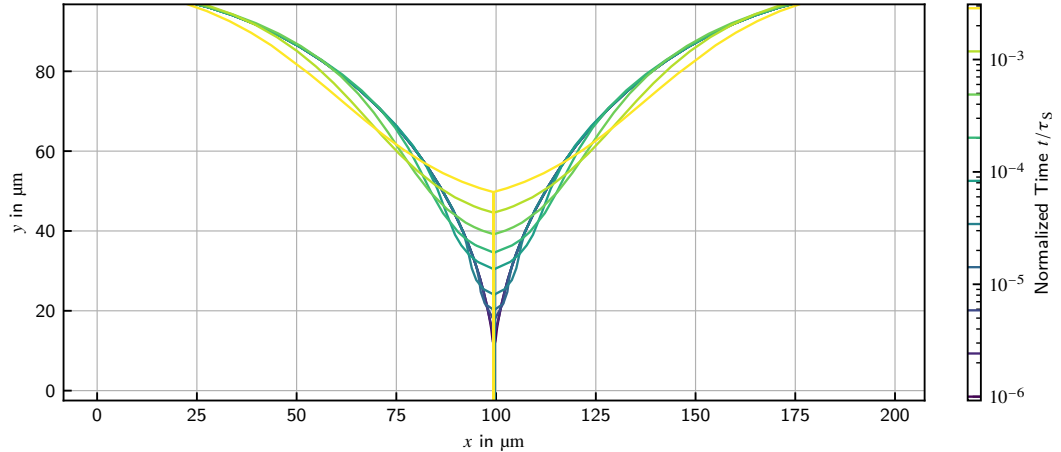


Appendix

Two-Particle Contacts

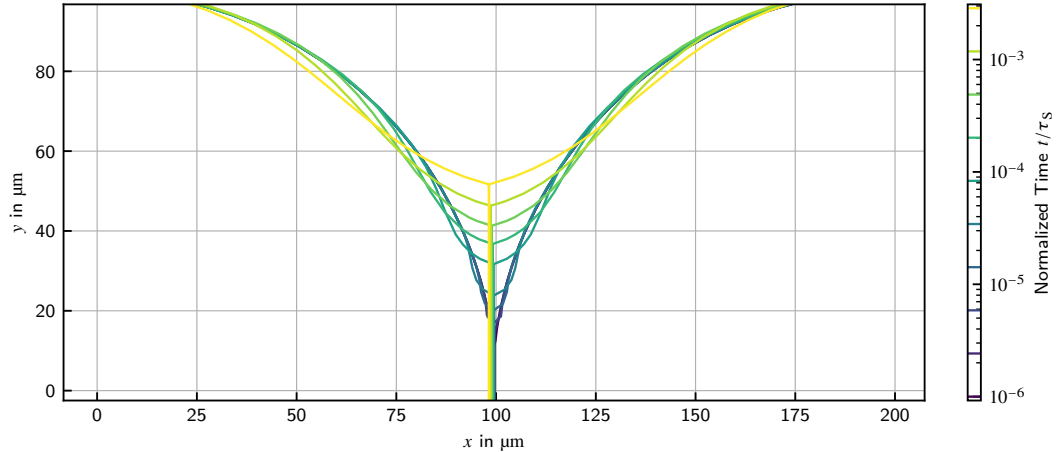
Surface-Boundary-Diffusion-Ratio

Geometry Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S



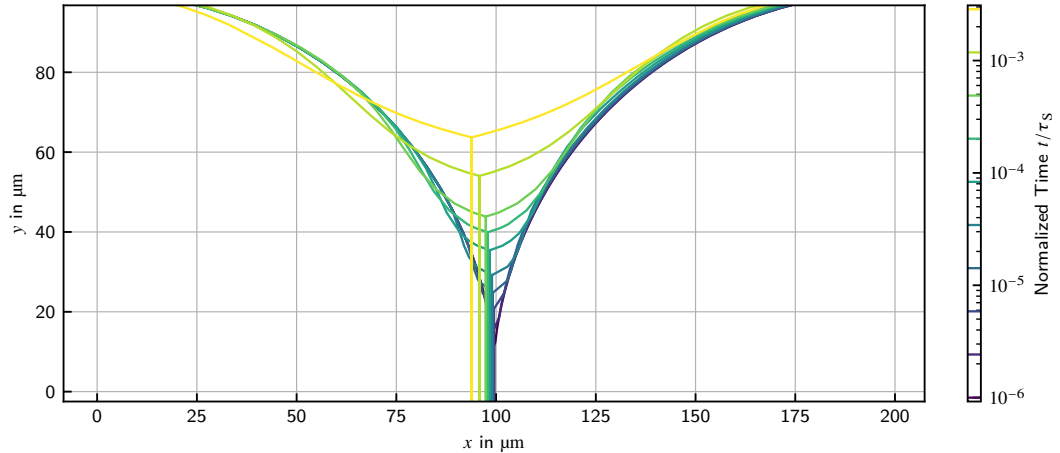
$$D_{GB}/D_S = 0.01$$

Geometry Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S



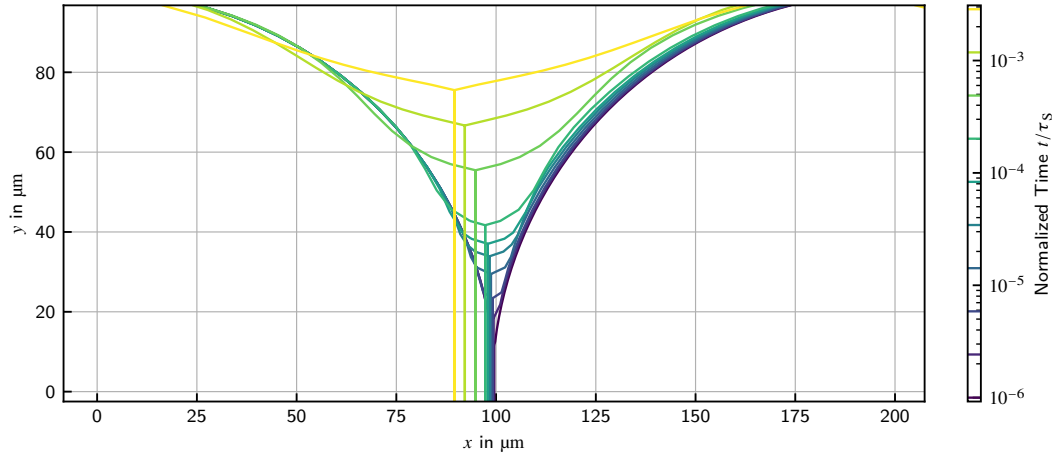
$$D_{GB}/D_S = 0.1$$

Geometry Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S



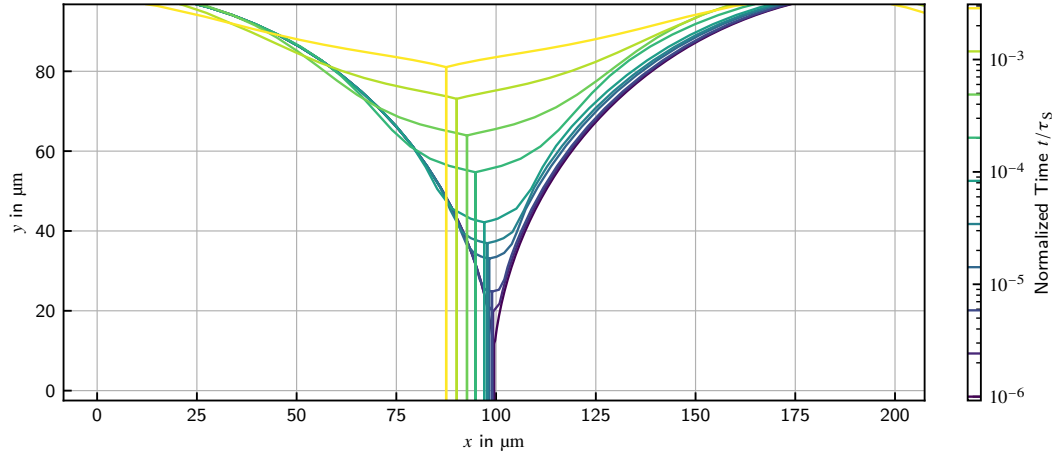
$$D_{GB}/D_S = 1$$

Geometry Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S



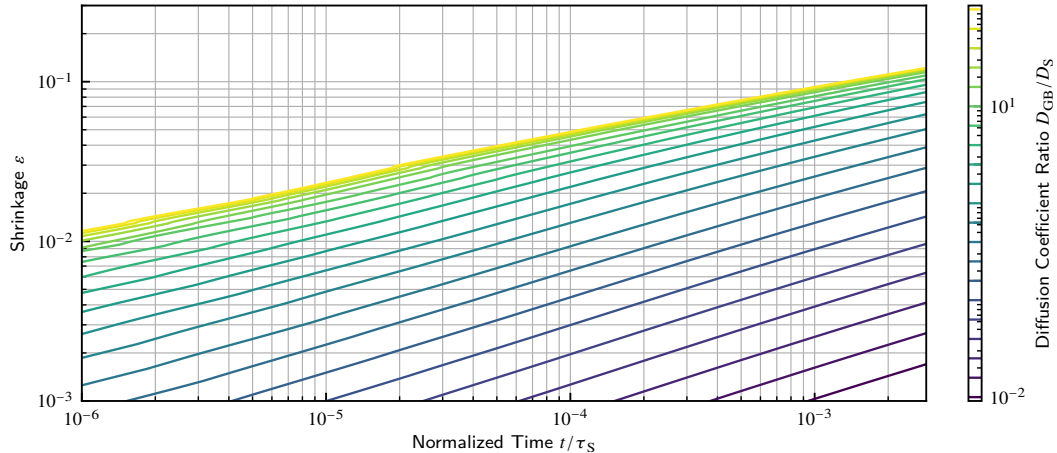
$$D_{GB}/D_S = 10$$

Geometry Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S

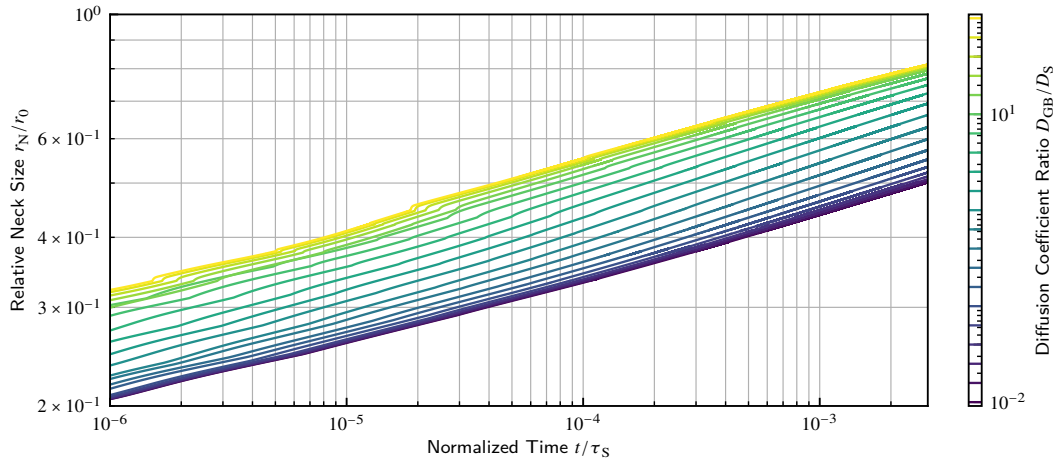


$$D_{GB}/D_S = 100$$

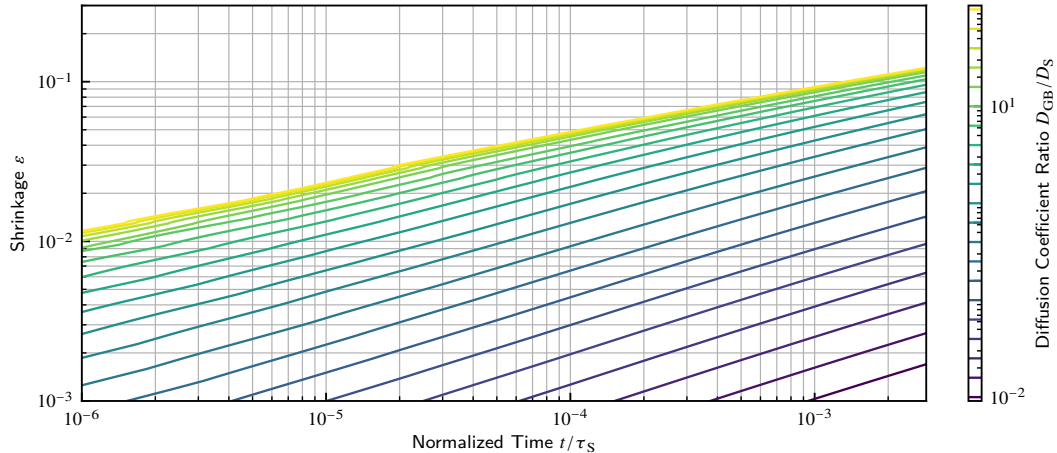
Shrinkage Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S



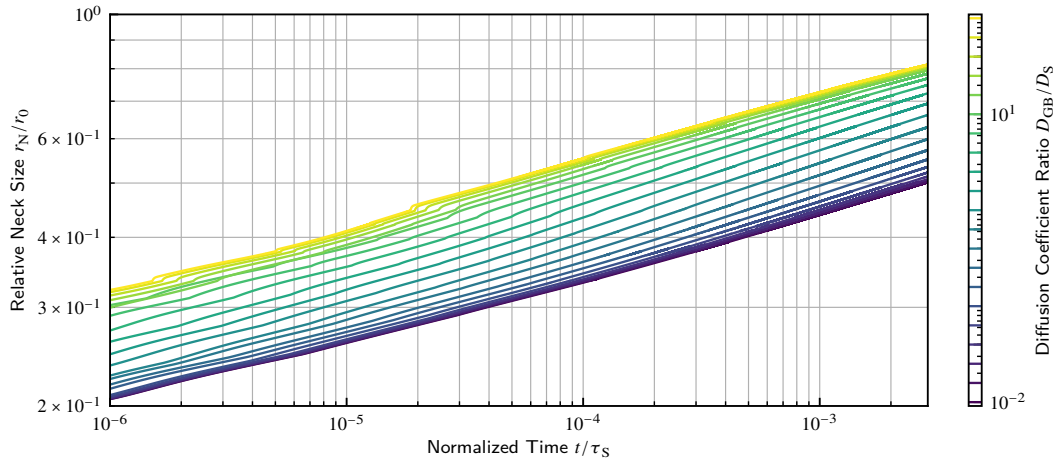
Neck Size Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S



Shrinkage Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S



Neck Size Evolution – Surface-Boundary-Diffusion-Ratio D_{GB}/D_S

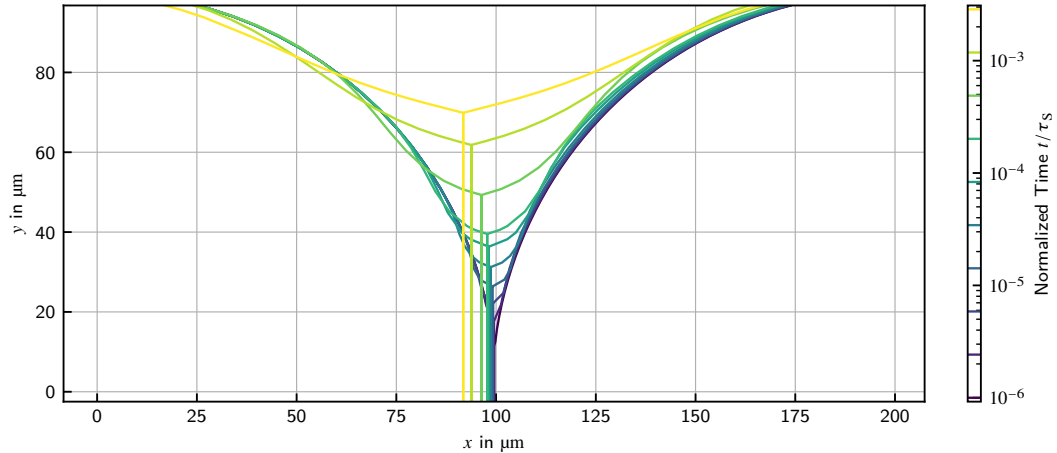


Appendix

Two-Particle Contacts

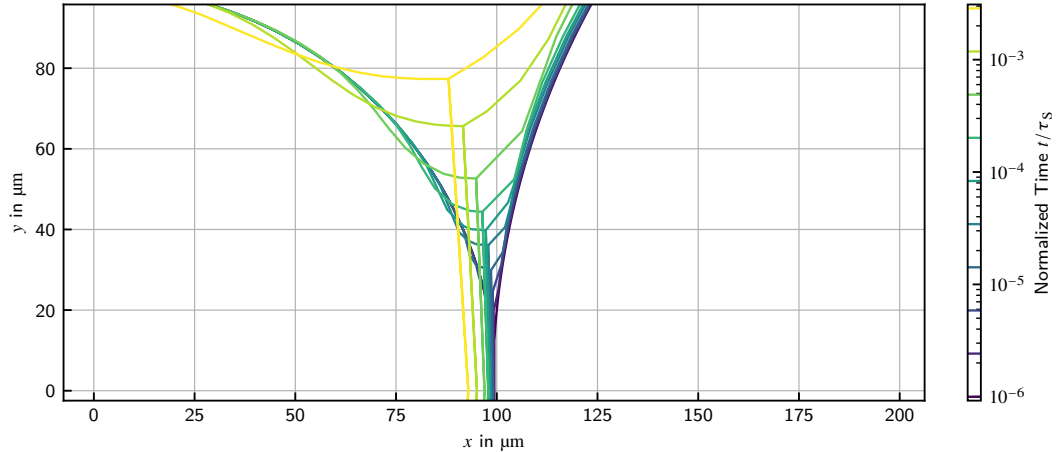
Particle-Size-Ratio

Geometry Evolution – Particle-Size-Ratio r_2/r_1

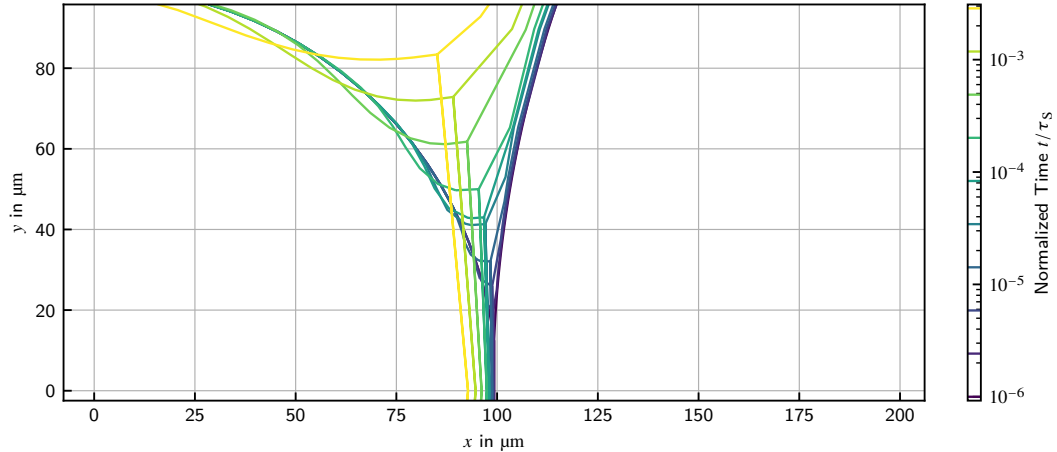


$$r_2/r_1 = 1$$

Geometry Evolution – Particle-Size-Ratio r_2/r_1

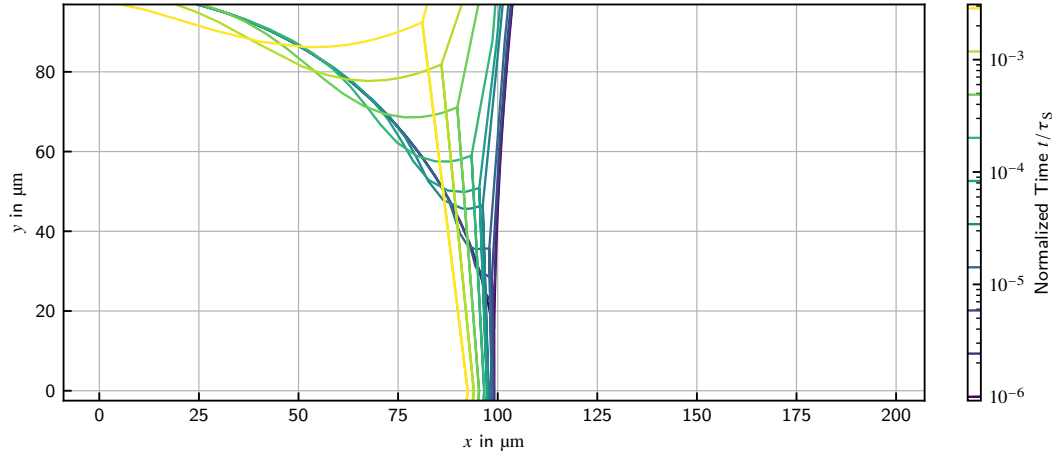


Geometry Evolution – Particle-Size-Ratio r_2/r_1

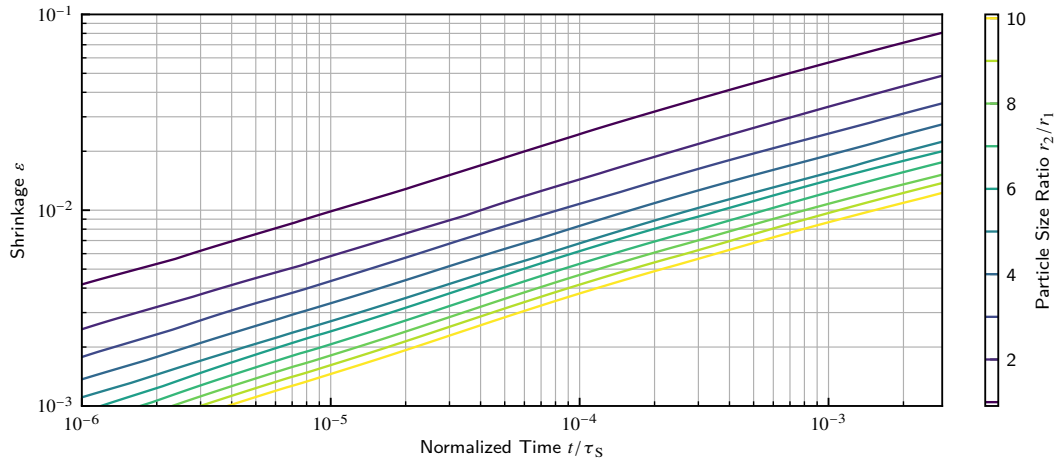


$$r_2/r_1 = 3$$

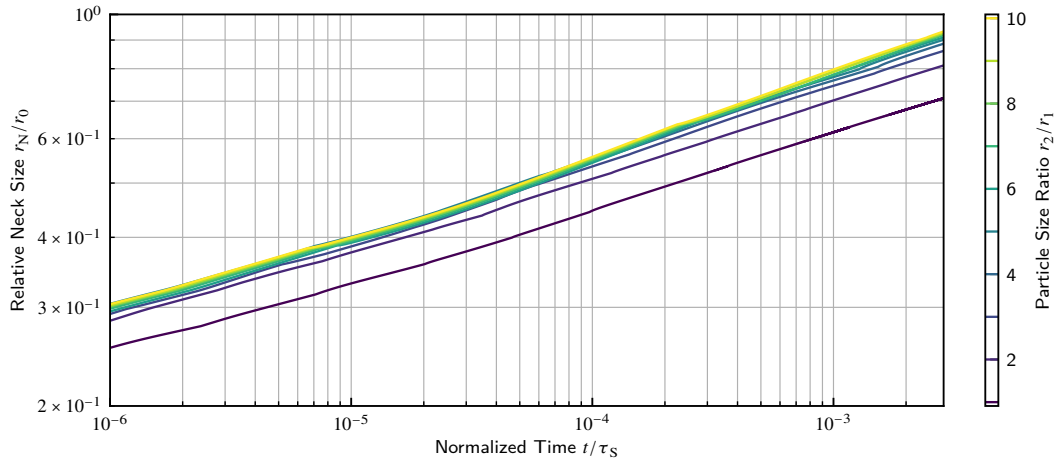
Geometry Evolution – Particle-Size-Ratio r_2/r_1



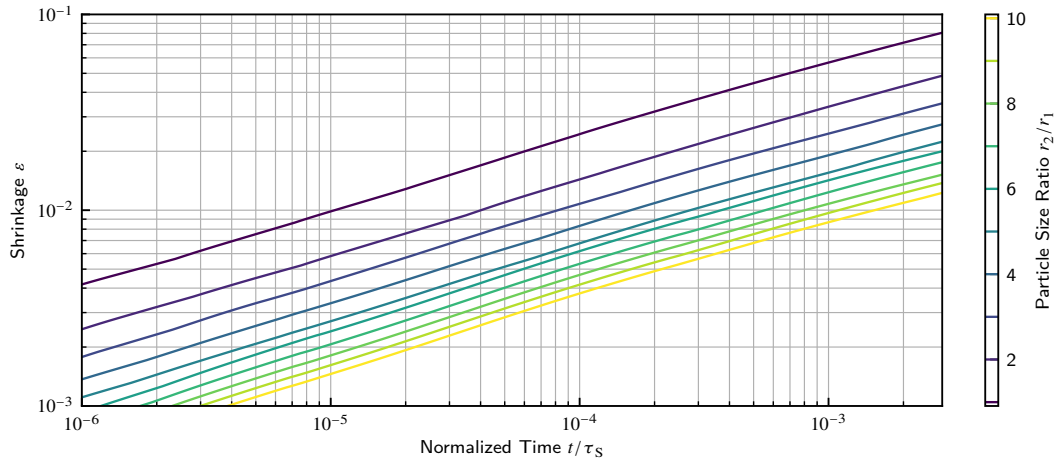
Shrinkage Evolution – Particle-Size-Ratio r_2/r_1



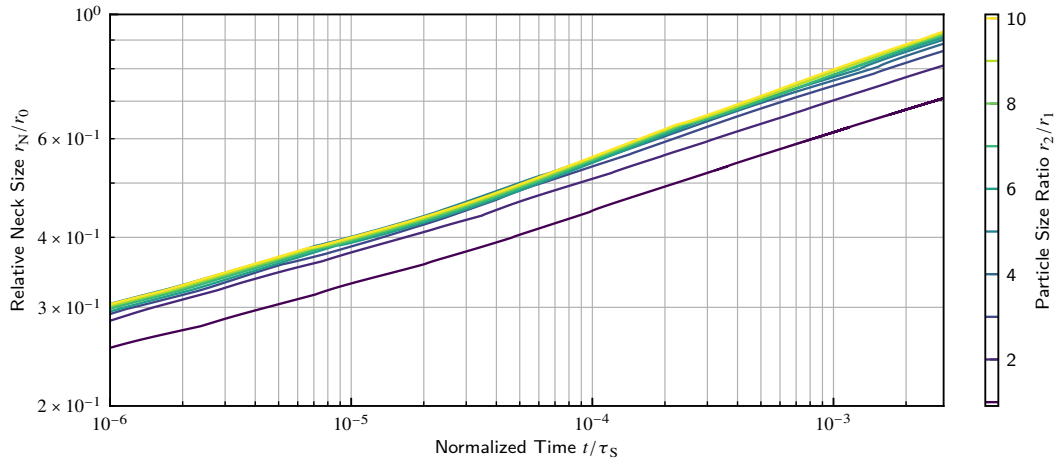
Neck Size Evolution – Particle-Size-Ratio r_2/r_1



Shrinkage Evolution – Particle-Size-Ratio r_2/r_1



Neck Size Evolution – Particle-Size-Ratio r_2/r_1

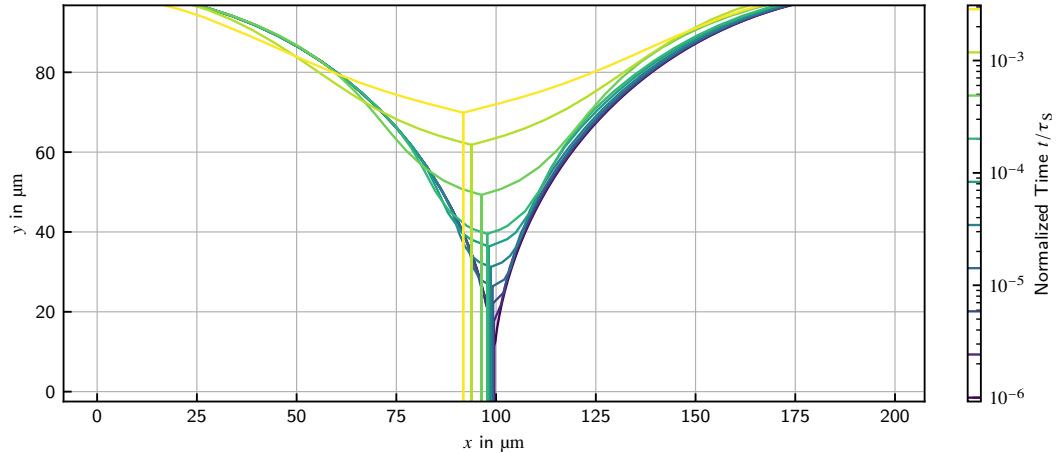


Appendix

Two-Particle Contacts

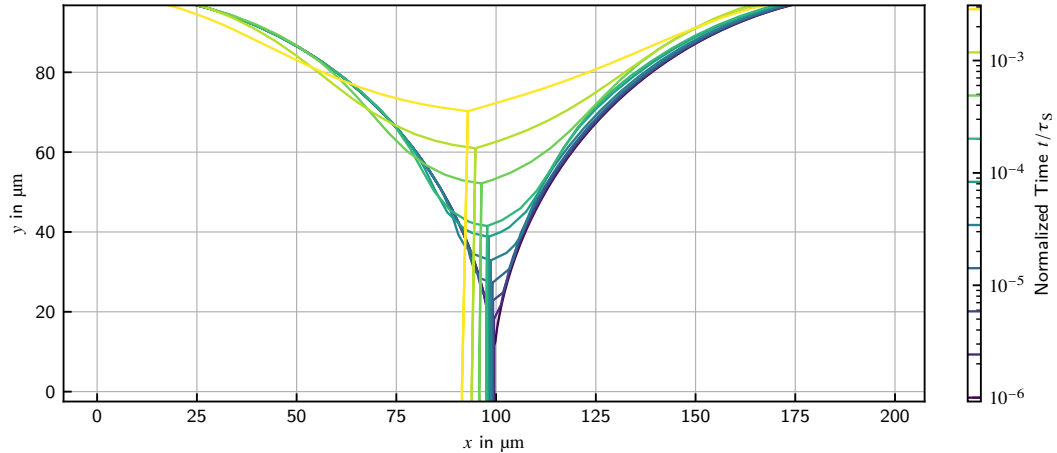
Surface-Energy-Asymmetry

Geometry Evolution – Surface-Energy-Asymmetry γ_{S2}/γ_{S1}

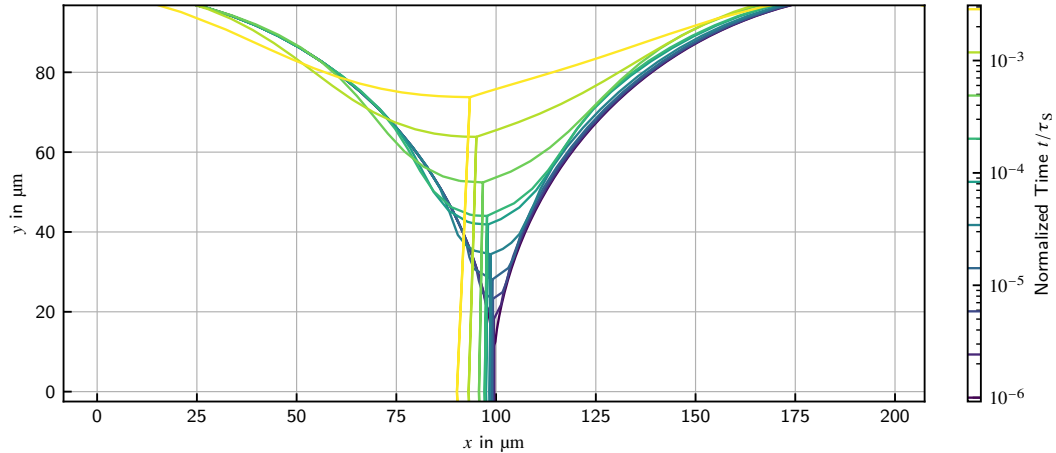


$$\gamma_{S2}/\gamma_{S1} = 1.0$$

Geometry Evolution – Surface-Energy-Asymmetry γ_{S2}/γ_{S1}

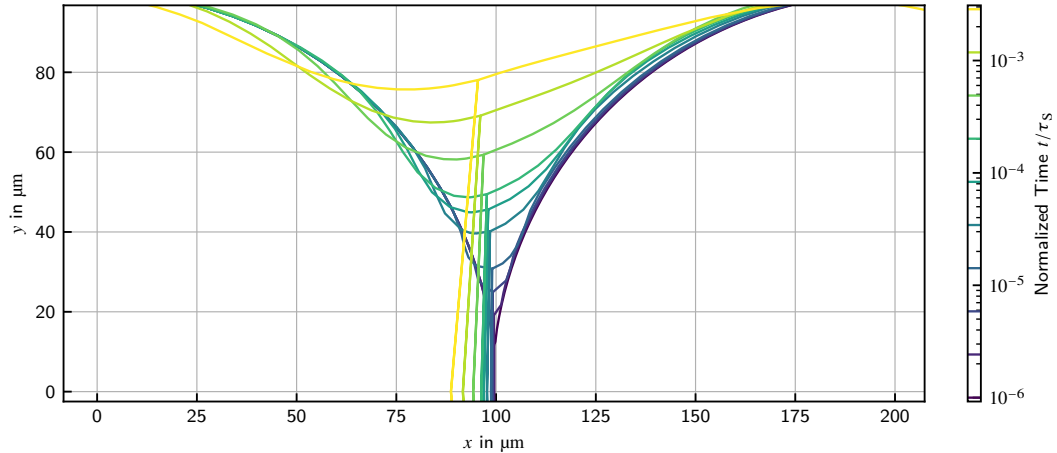


Geometry Evolution – Surface-Energy-Asymmetry γ_{S2}/γ_{S1}

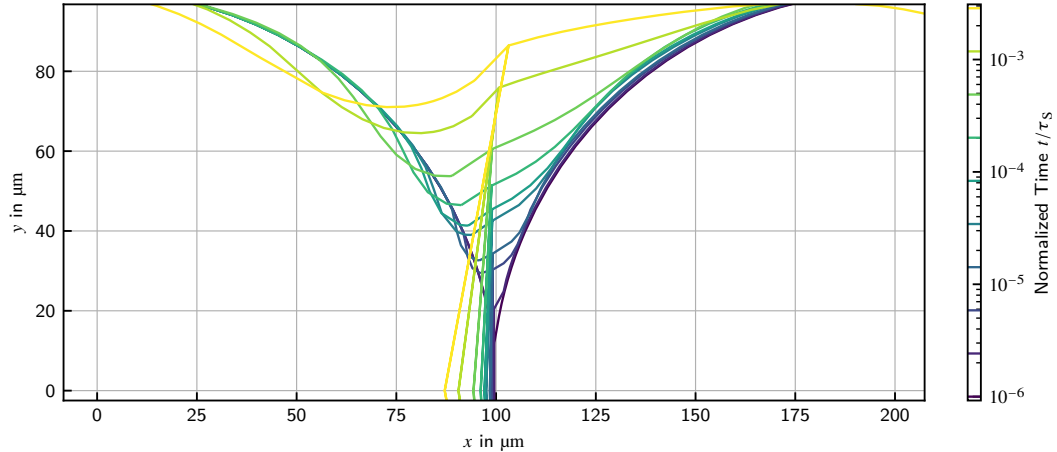


$$\gamma_{S2}/\gamma_{S1} = 1.5$$

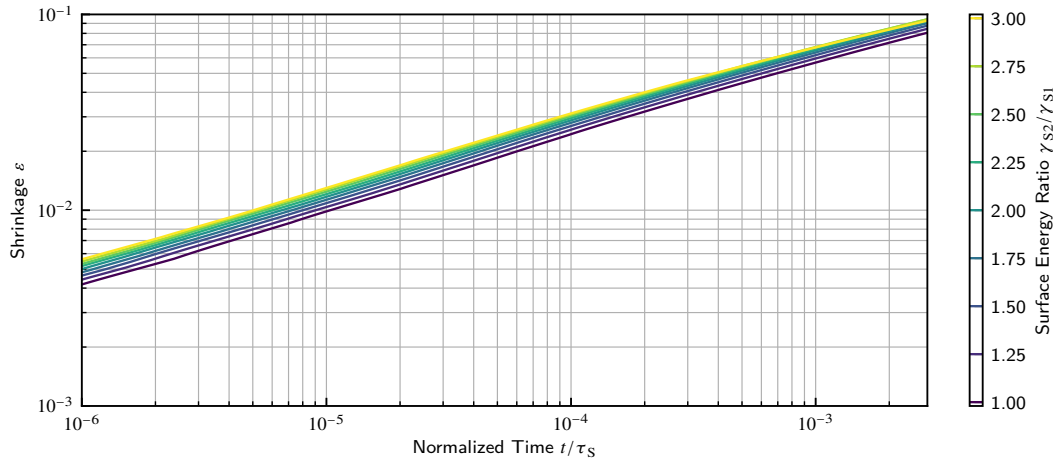
Geometry Evolution – Surface-Energy-Asymmetry γ_{S2}/γ_{S1}



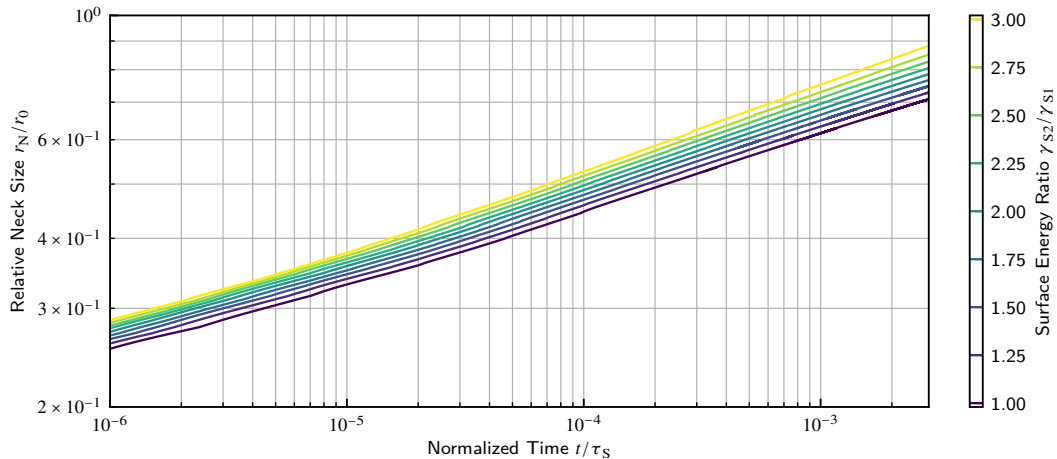
Geometry Evolution – Surface-Energy-Asymmetry γ_{S2}/γ_{S1}



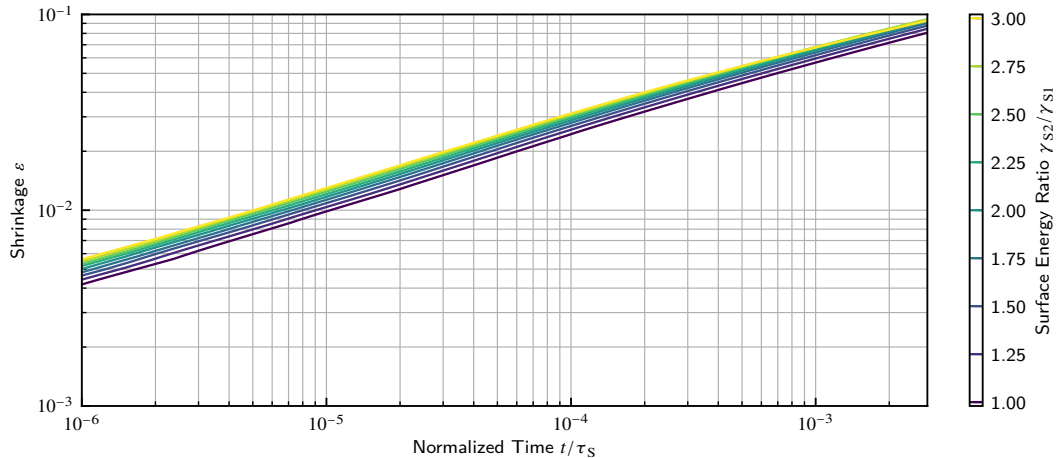
Shrinkage Evolution – Surface-Energy-Asymmetry γ_{s2}/γ_{s1}



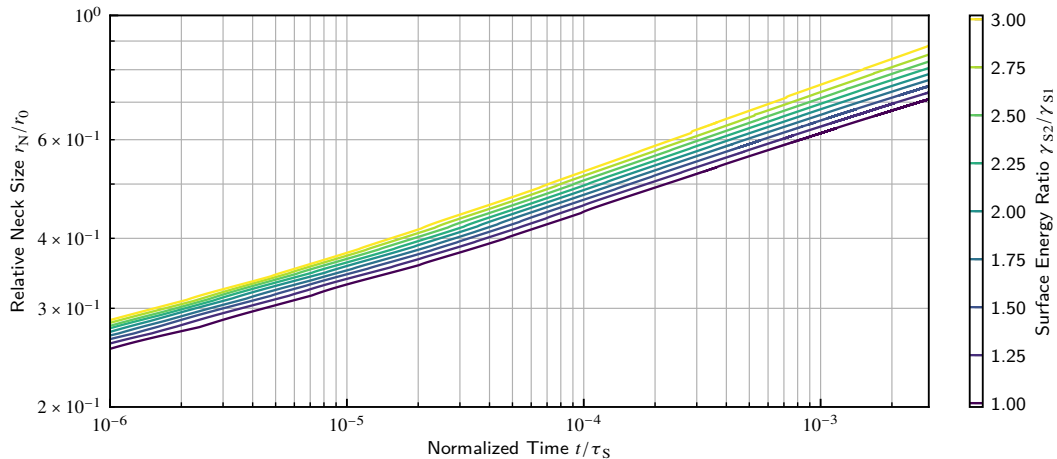
Neck Size Evolution – Surface-Energy-Asymmetry γ_{s2}/γ_{s1}



Shrinkage Evolution – Surface-Energy-Asymmetry γ_{s2}/γ_{s1}



Neck Size Evolution – Surface-Energy-Asymmetry γ_{s2}/γ_{s1}

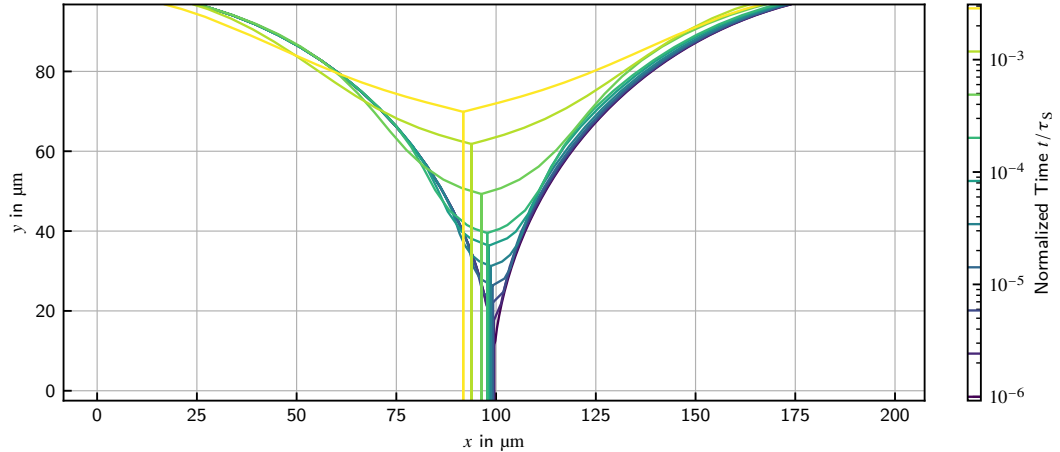


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Two-Particle Contacts

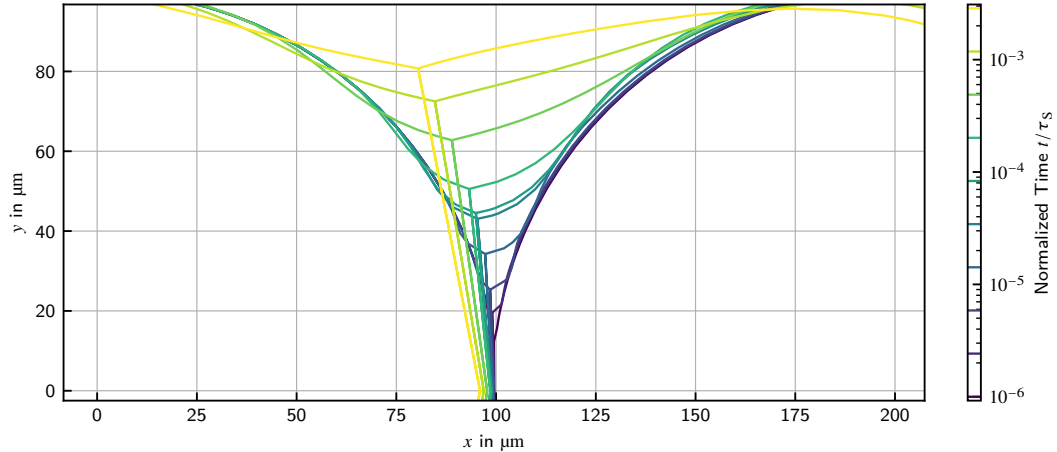
Diffusion-Asymmetry

Geometry Evolution – Diffusion-Asymmetry D_2/D_1



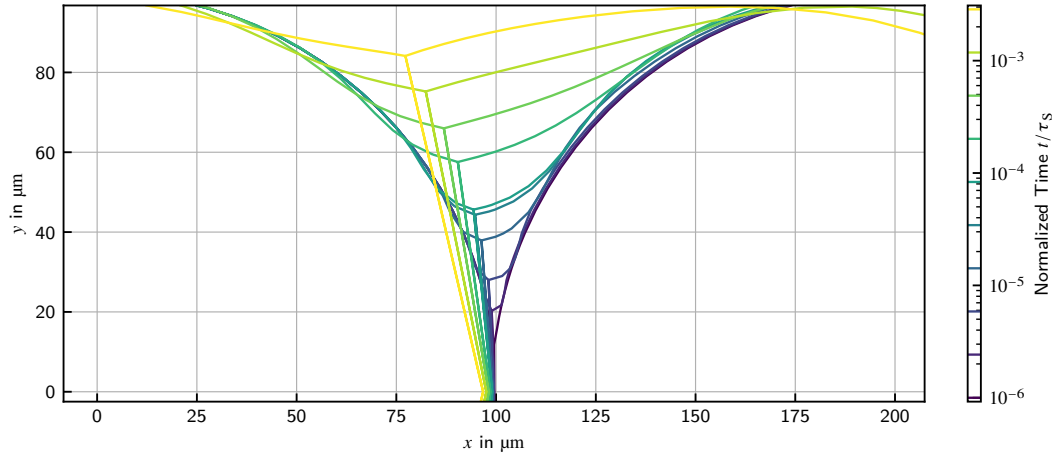
$$D_2/D_1 = 1.0$$

Geometry Evolution – Diffusion-Asymmetry D_2/D_1

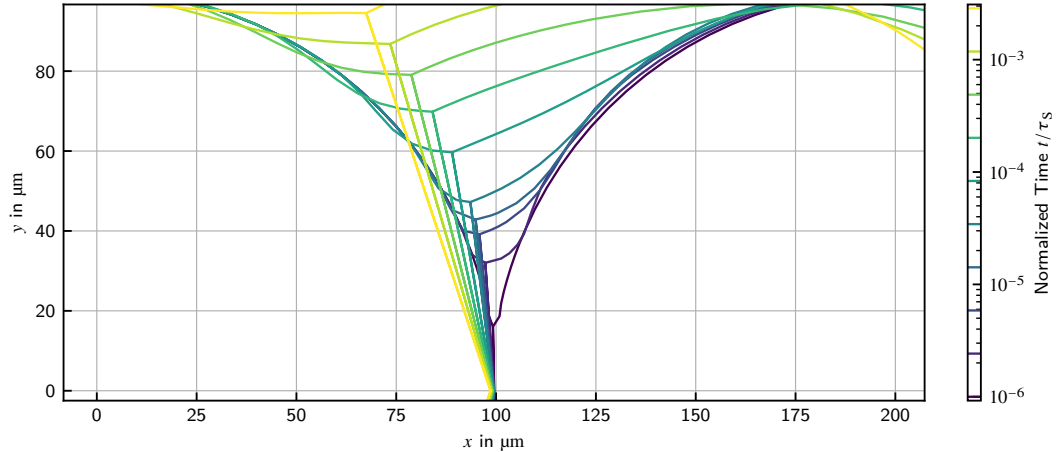


$$D_2/D_1 = 5.0$$

Geometry Evolution – Diffusion-Asymmetry D_2/D_1

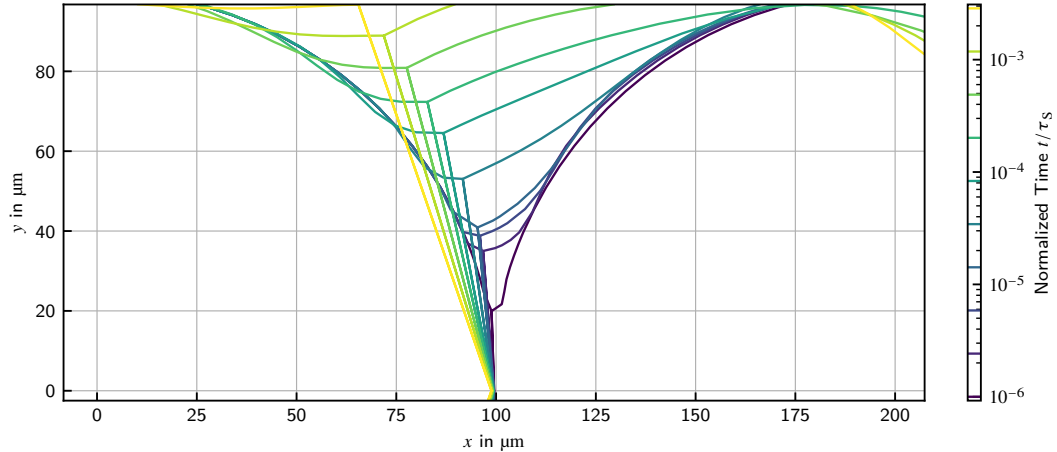


Geometry Evolution – Diffusion-Asymmetry D_2/D_1



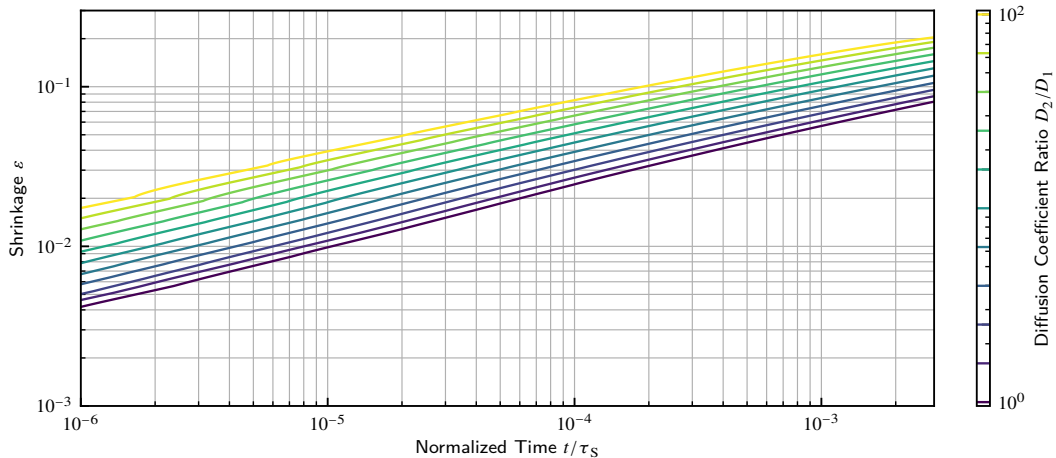
$$D_2/D_1 = 50.0$$

Geometry Evolution – Diffusion-Asymmetry D_2/D_1

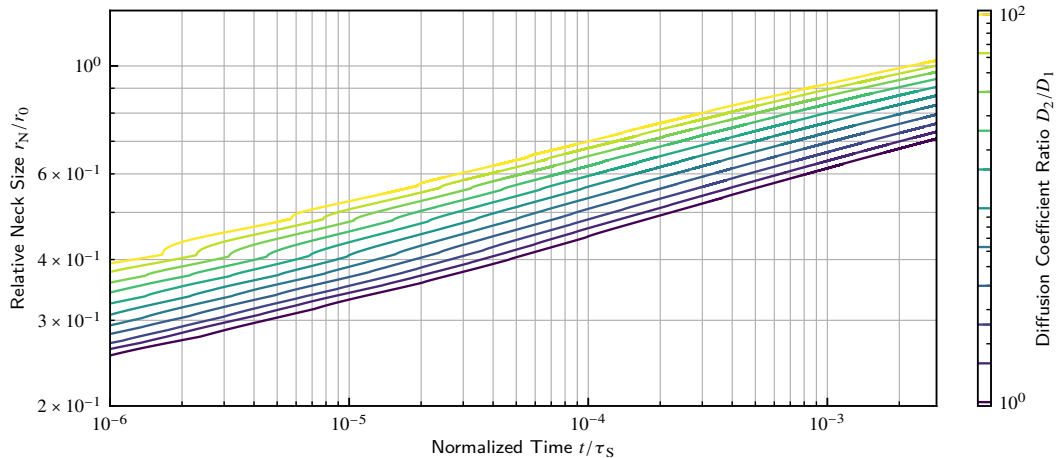


$$D_2/D_1 = 100.0$$

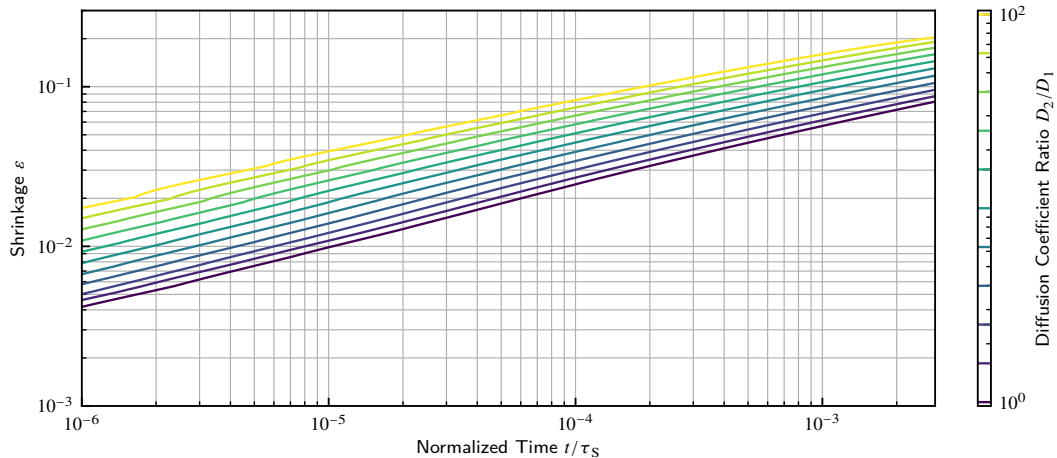
Shrinkage Evolution – Diffusion-Asymmetry D_2/D_1



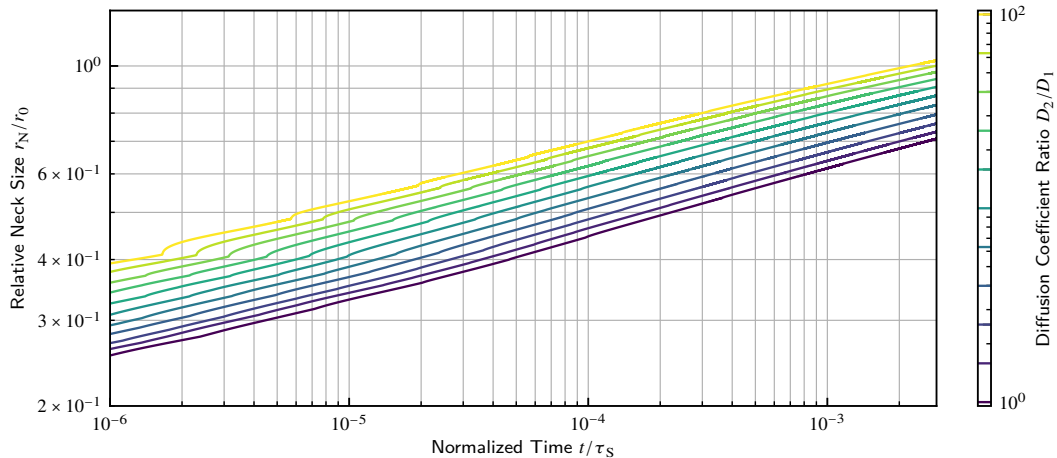
Neck Size Evolution – Diffusion-Asymmetry D_2/D_1



Shrinkage Evolution – Diffusion-Asymmetry D_2/D_1



Neck Size Evolution – Diffusion-Asymmetry D_2/D_1

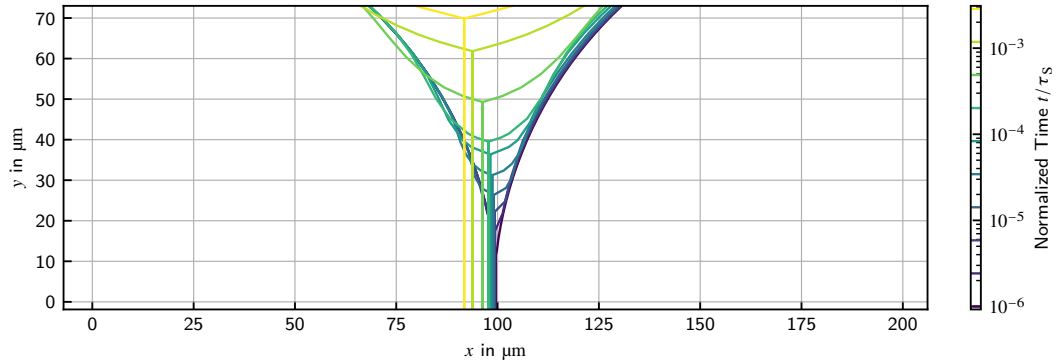


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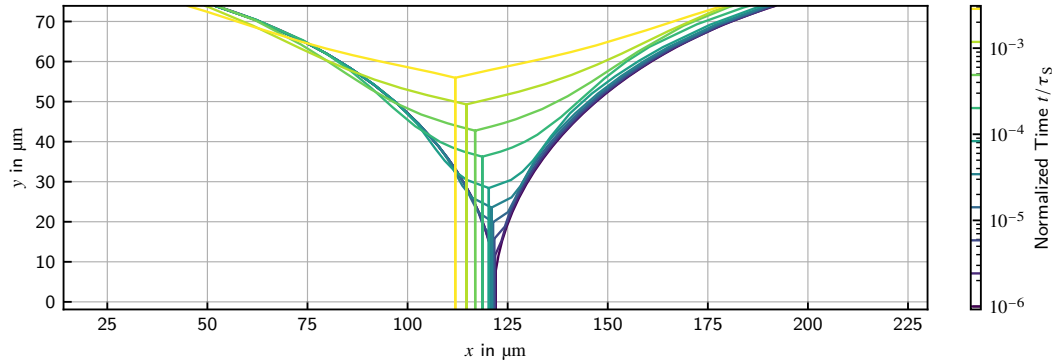
Two-Particle Contacts

Ovality – Tip-Tip-Configuration

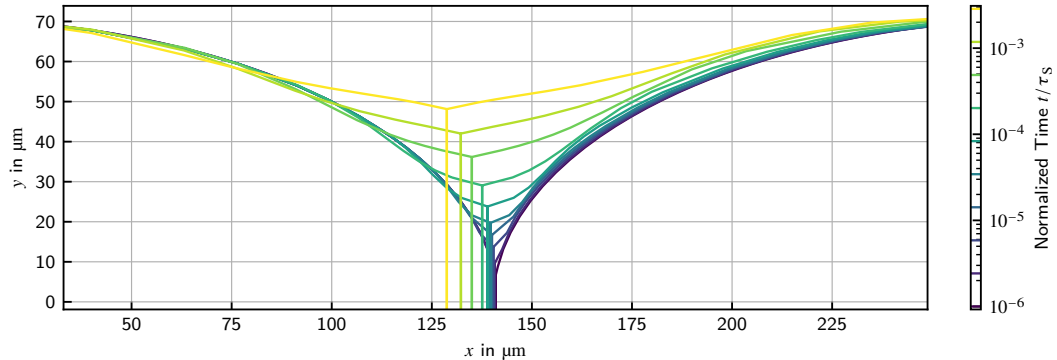
Geometry Evolution – Ovality o – Tip-Tip-Configuration



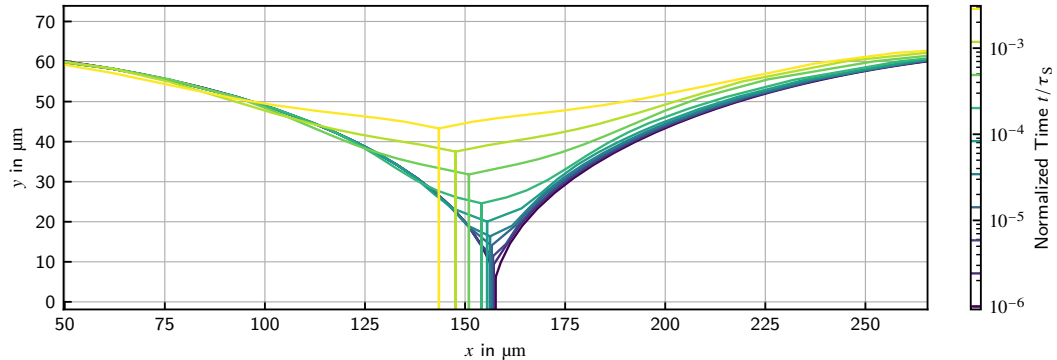
Geometry Evolution – Ovality o – Tip-Tip-Configuration



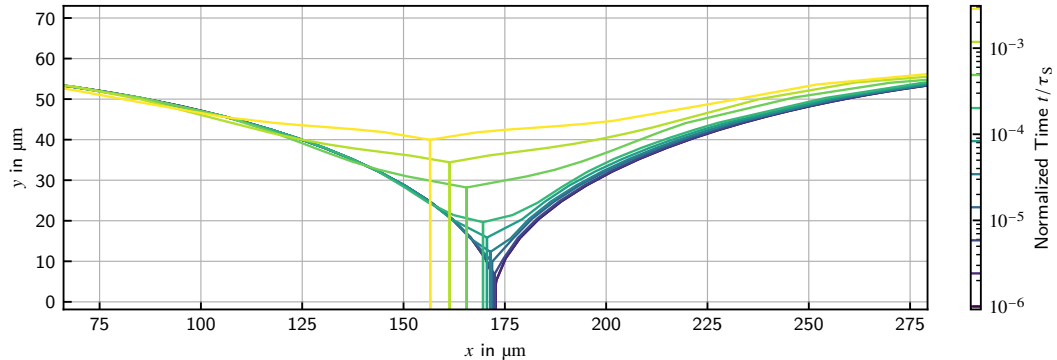
Geometry Evolution – Ovality o – Tip-Tip-Configuration



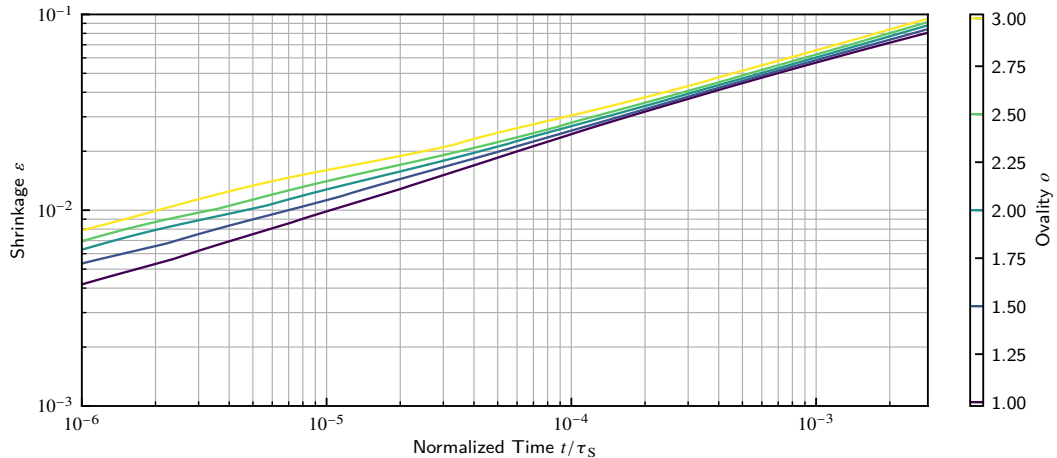
Geometry Evolution – Ovality o – Tip-Tip-Configuration



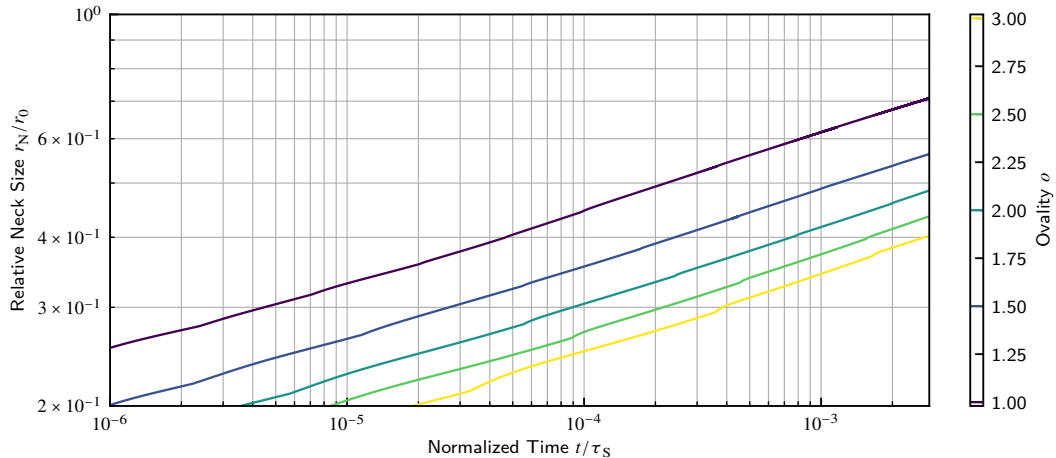
Geometry Evolution – Ovality o – Tip-Tip-Configuration



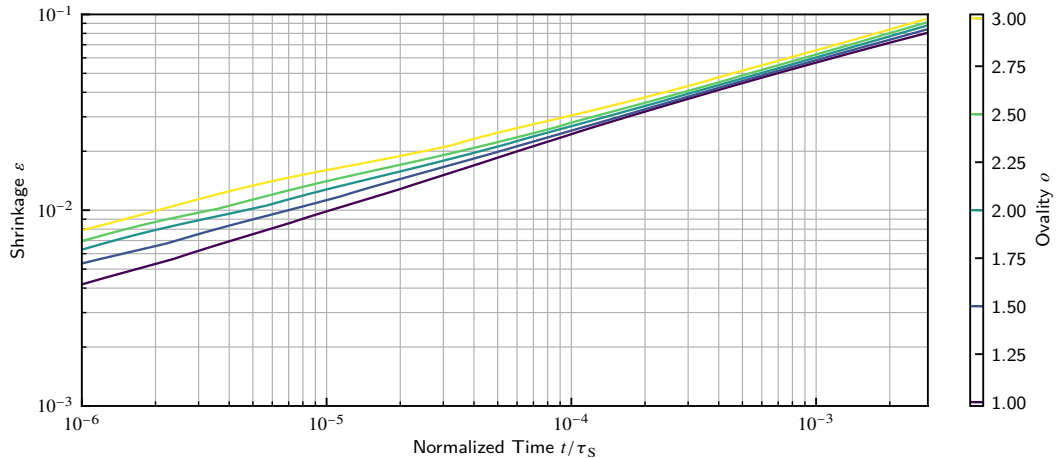
Shrinkage Evolution – Ovality o – Tip-Tip-Configuration



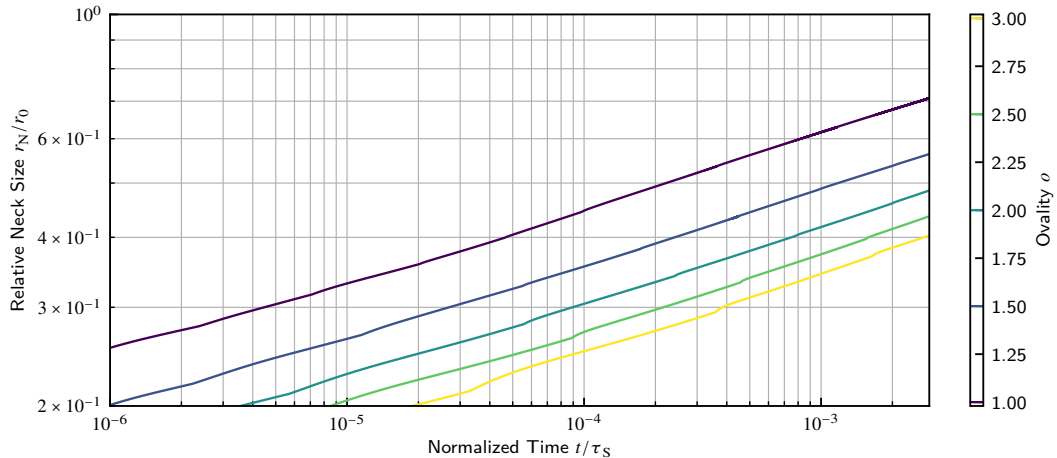
Neck Size Evolution – Ovality o – Tip-Tip-Configuration



Shrinkage Evolution – Ovality o – Tip-Tip-Configuration



Neck Size Evolution – Ovality o – Tip-Tip-Configuration

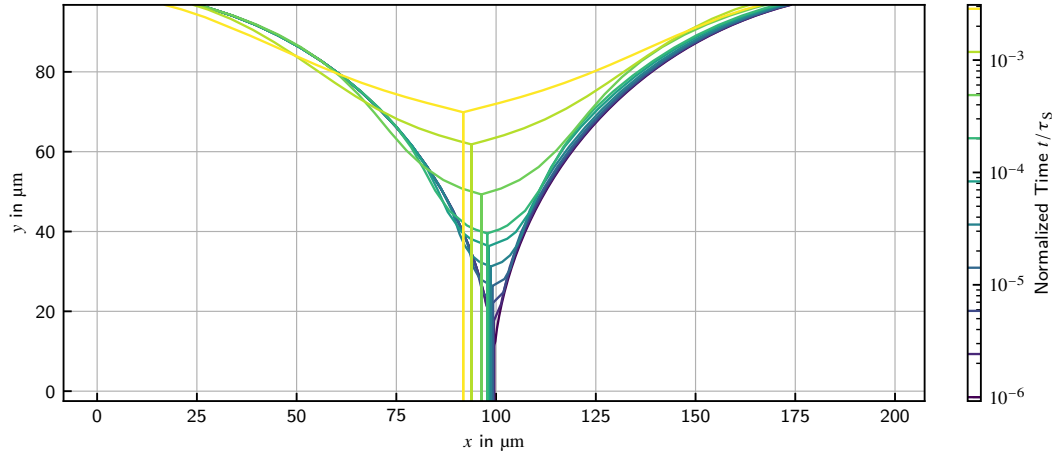


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Two-Particle Contacts

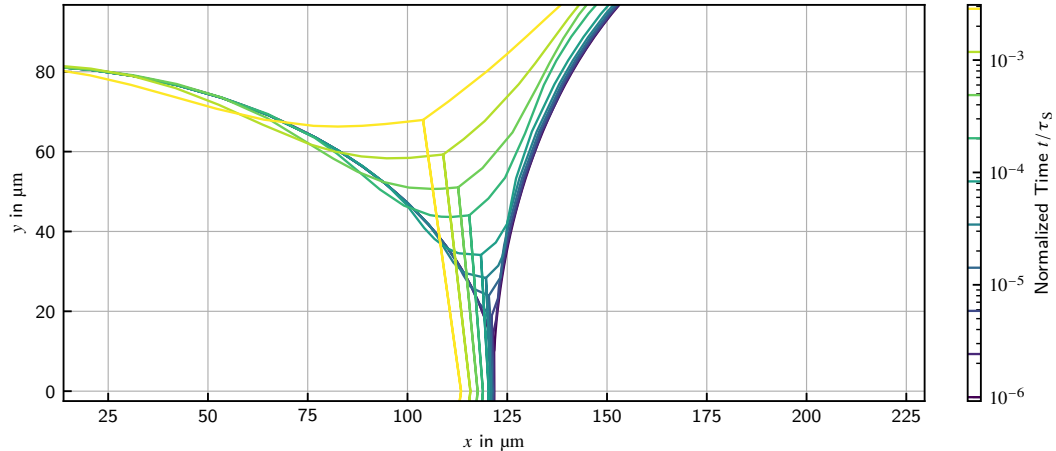
Ovality – Tip-Flank-Configuration

Geometry Evolution – Ovality o – Tip-Flank-Configuration



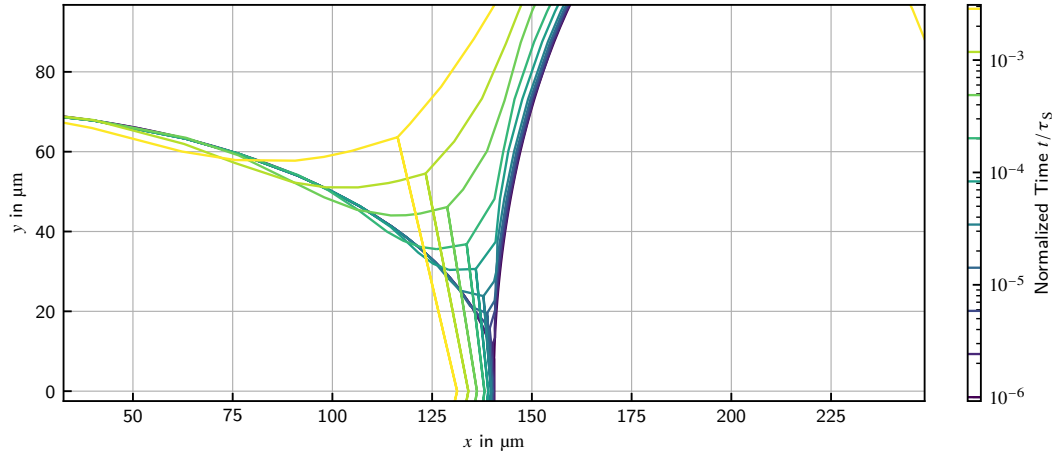
$$o = 1.0$$

Geometry Evolution – Ovality o – Tip-Flank-Configuration



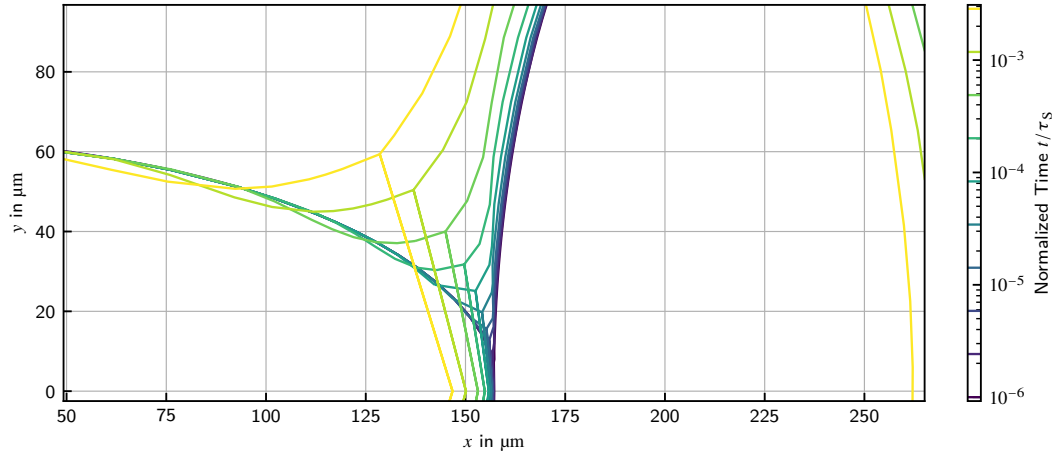
$$o = 1.5$$

Geometry Evolution – Ovality o – Tip-Flank-Configuration



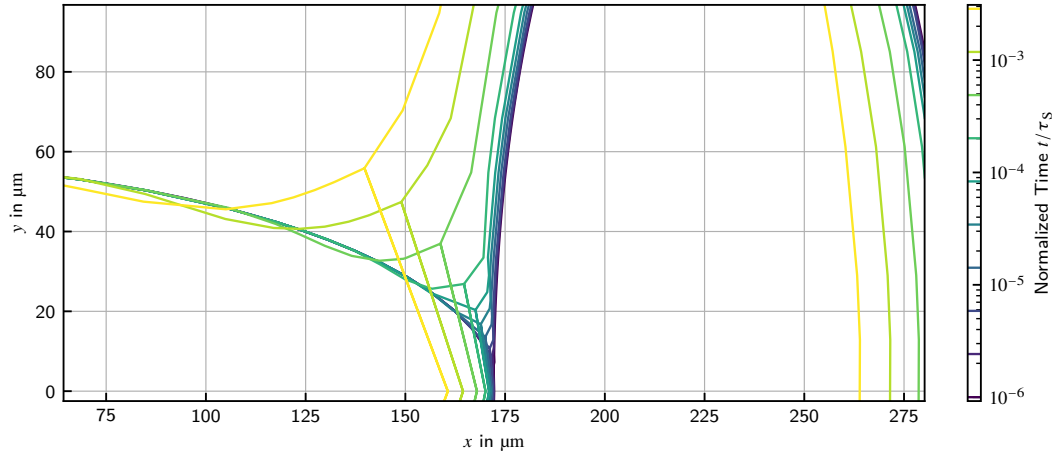
$$o = 2.0$$

Geometry Evolution – Ovality o – Tip-Flank-Configuration



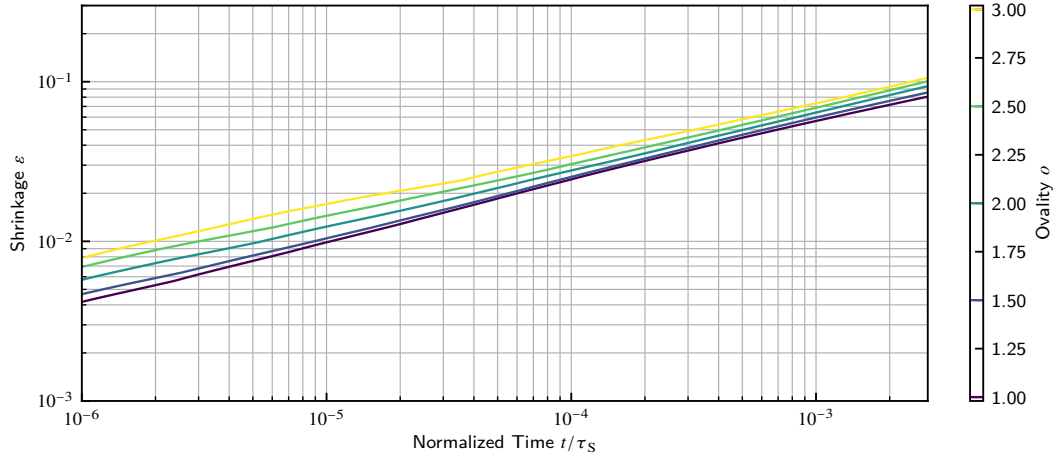
$$o = 2.5$$

Geometry Evolution – Ovality o – Tip-Flank-Configuration

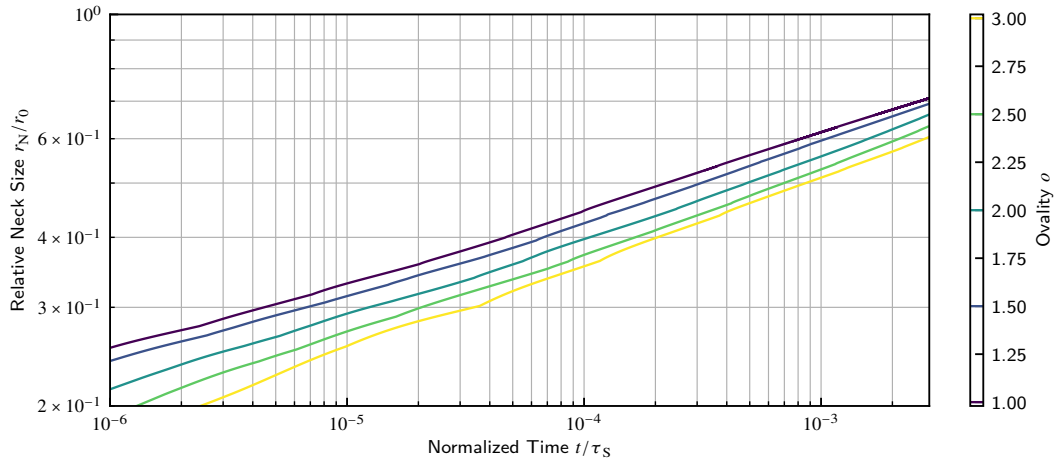


$$o = 3.0$$

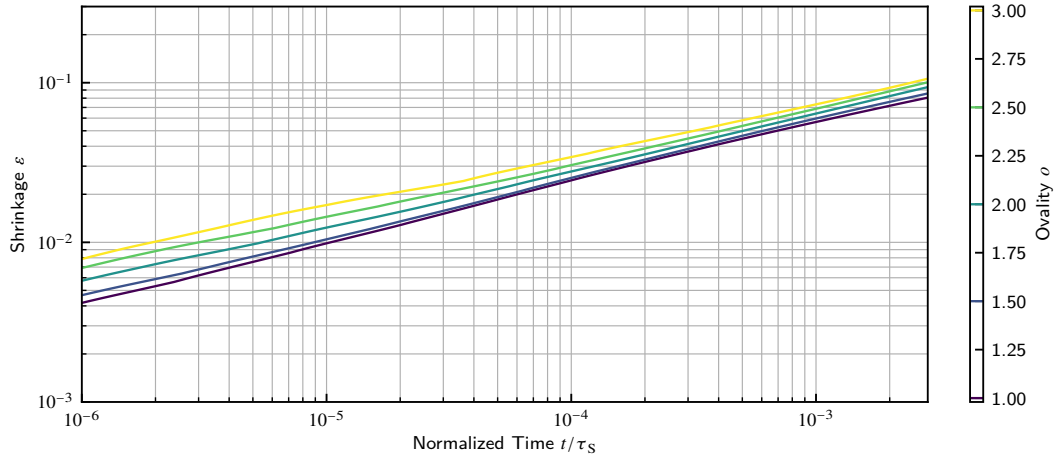
Shrinkage Evolution – Ovality o – Tip-Flank-Configuration



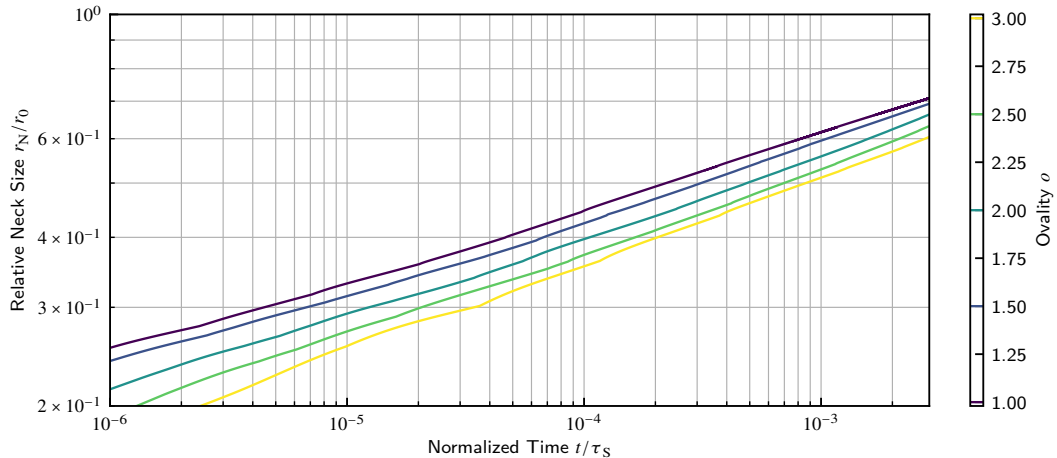
Neck Size Evolution – Ovality o – Tip-Flank-Configuration



Shrinkage Evolution – Ovality o – Tip-Flank-Configuration



Neck Size Evolution – Ovality o – Tip-Flank-Configuration

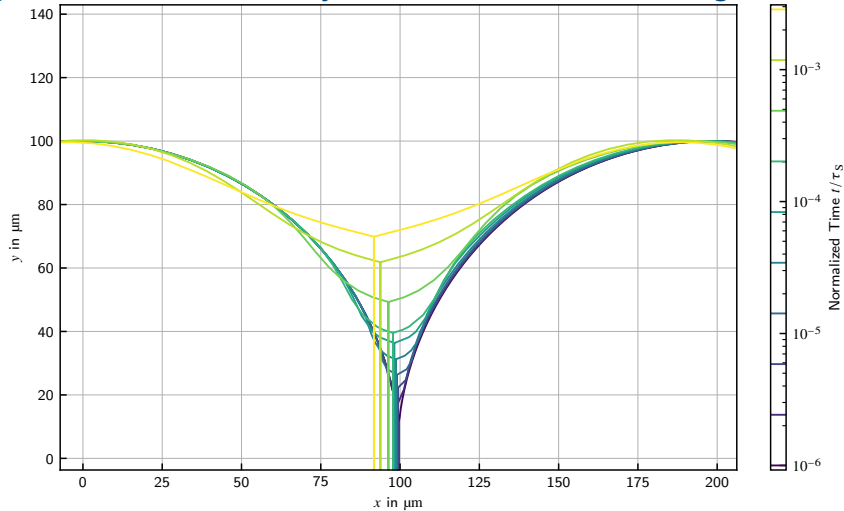


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Two-Particle Contacts

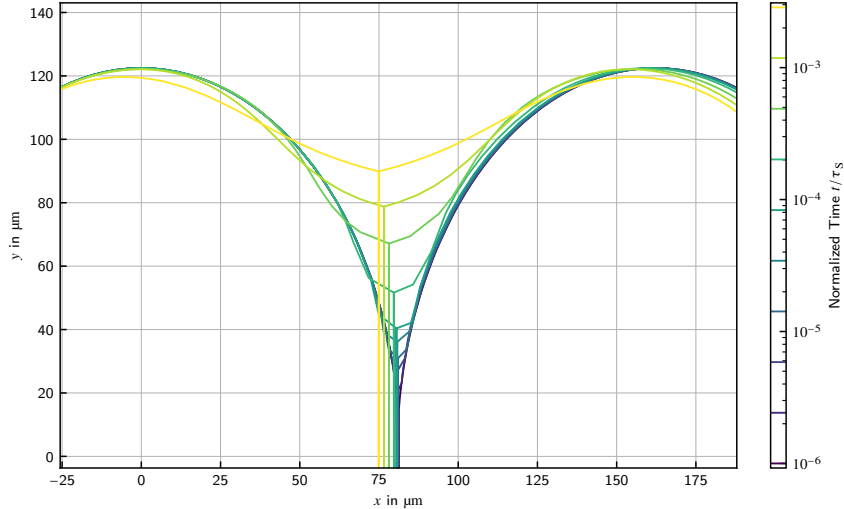
Ovality – Flank-Flank-Configuration

Geometry Evolution – Ovality o – Flank-Flank-Configuration



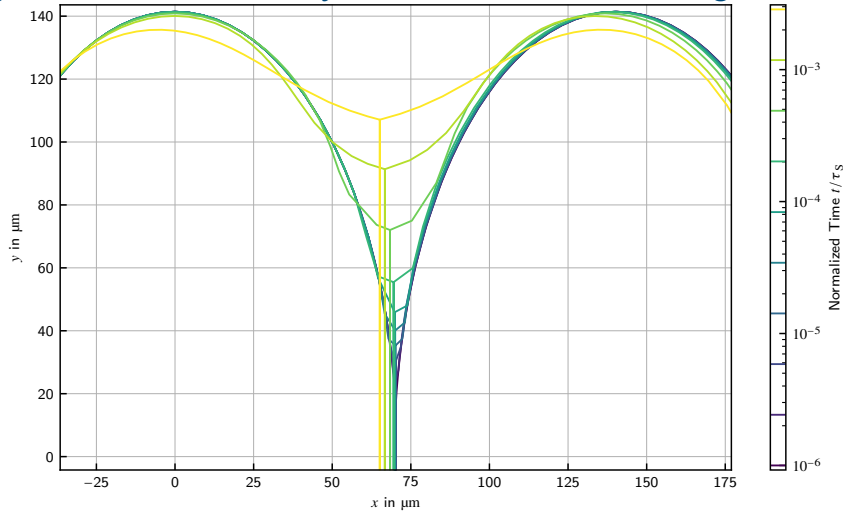
$$o = 1.0$$

Geometry Evolution – Ovality o – Flank-Flank-Configuration



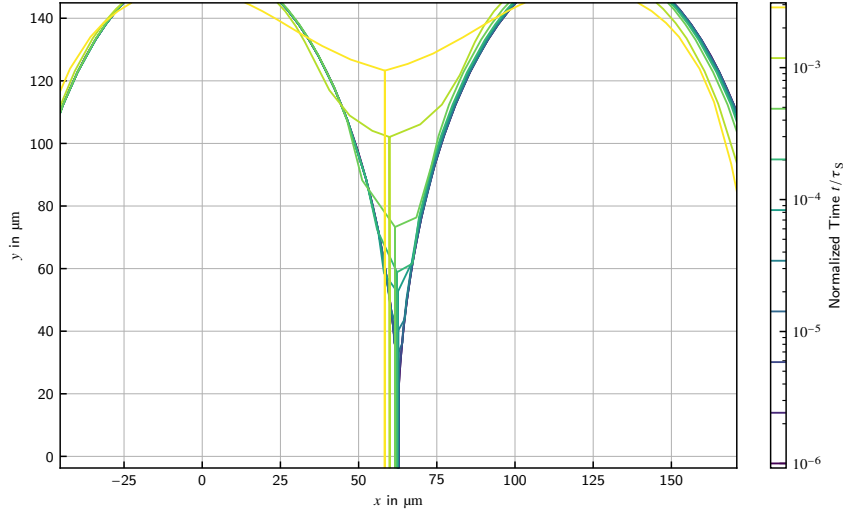
$$o = 1.5$$

Geometry Evolution – Ovality o – Flank-Flank-Configuration



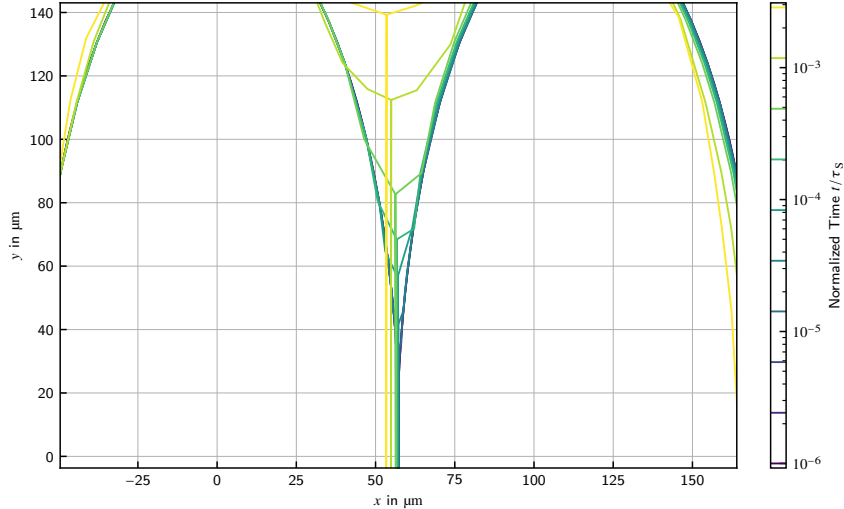
$$o = 2.0$$

Geometry Evolution – Ovality o – Flank-Flank-Configuration



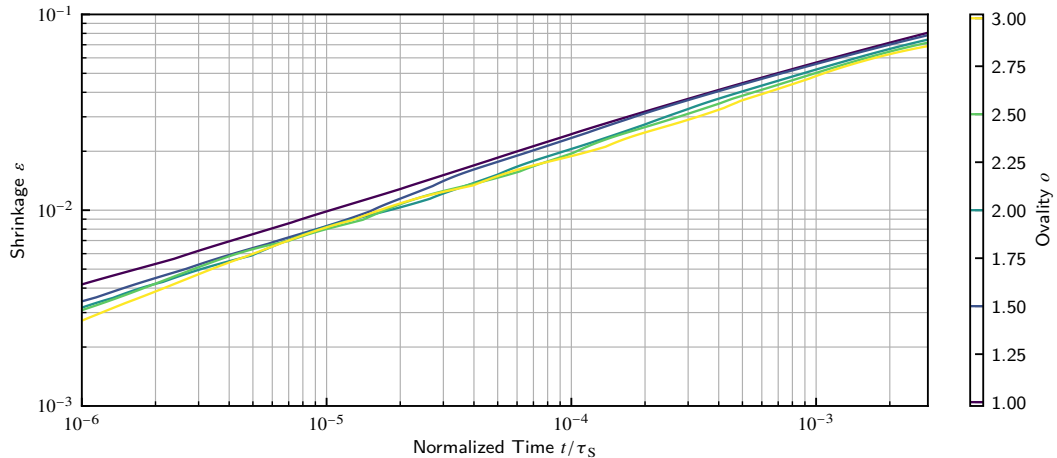
$$o = 2.5$$

Geometry Evolution – Ovality o – Flank-Flank-Configuration

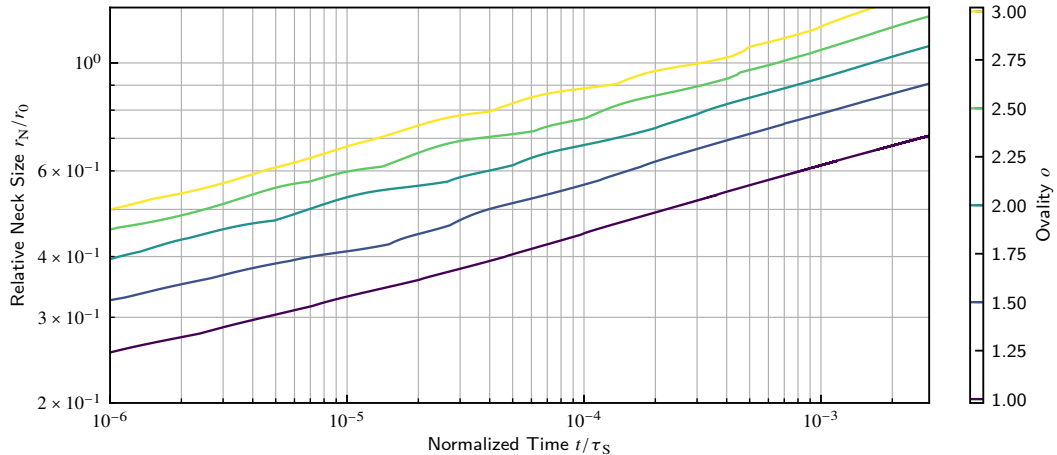


$$o = 3.0$$

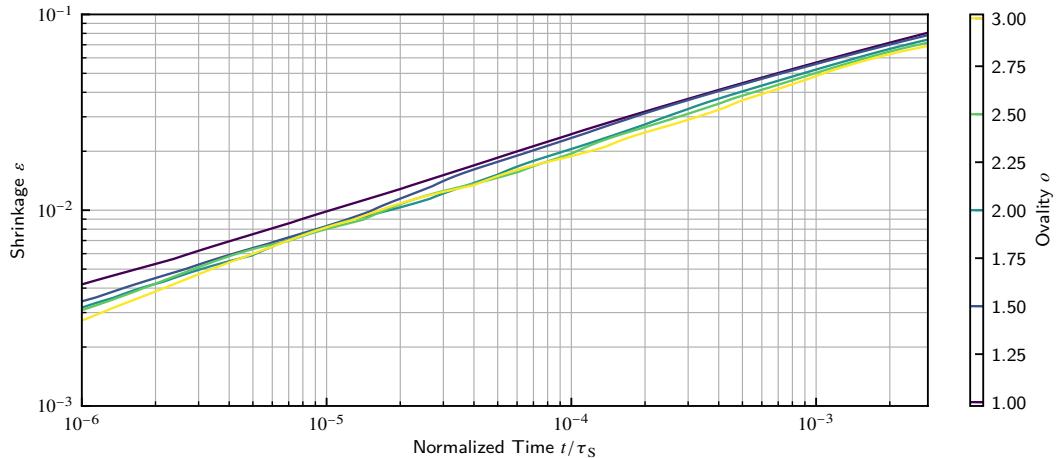
Shrinkage Evolution – Ovality o – Flank-Flank-Configuration



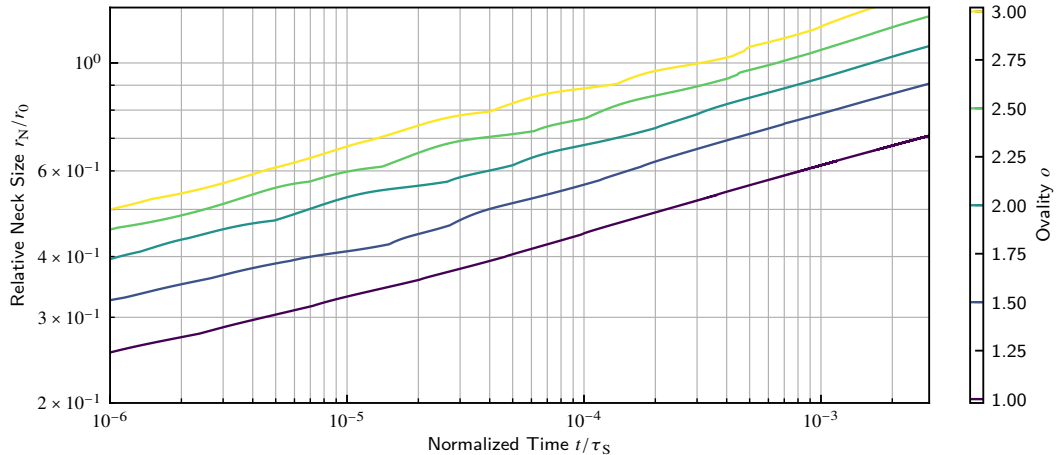
Neck Size Evolution – Ovality o – Flank-Flank-Configuration



Shrinkage Evolution – Ovality o – Flank-Flank-Configuration



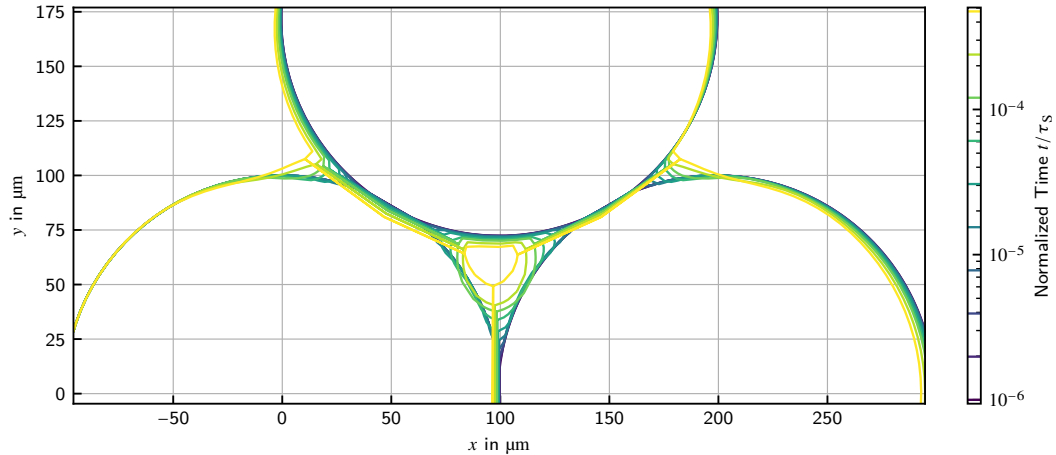
Neck Size Evolution – Ovality o – Flank-Flank-Configuration



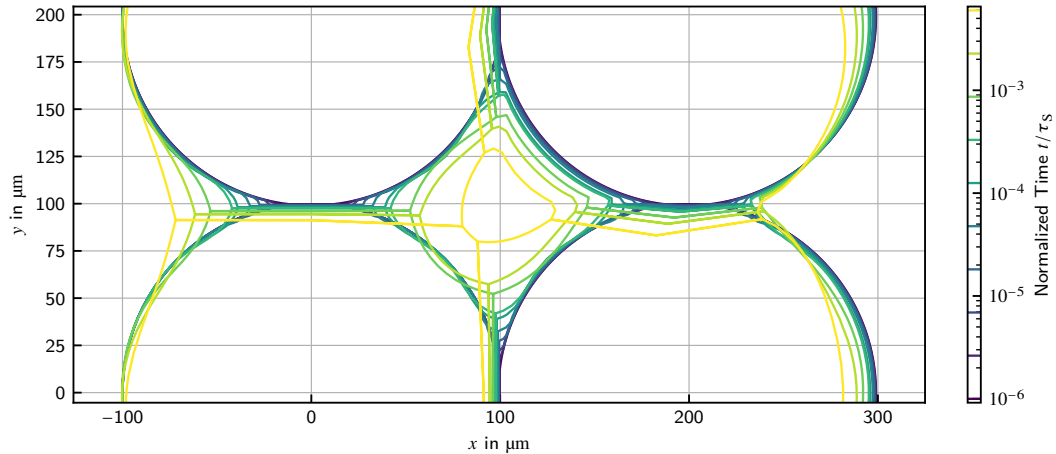
Appendix

Multi-Particle Contacts

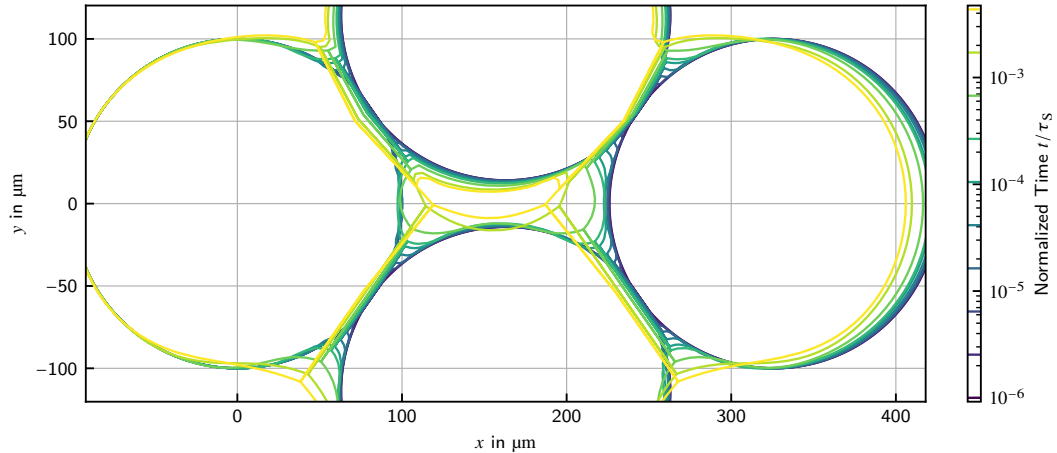
Geometry Evolution - Triangle Packing With Inert



Geometry Evolution - Square Packing With Inert



Geometry Evolution - Rhombus Packing With Inert



Appendix

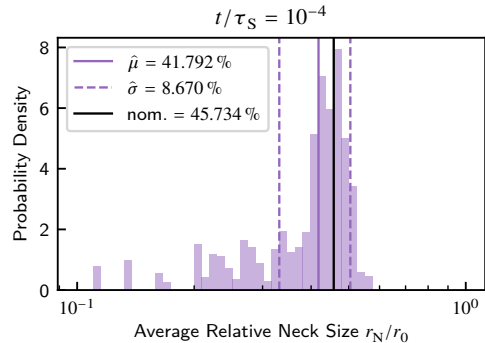
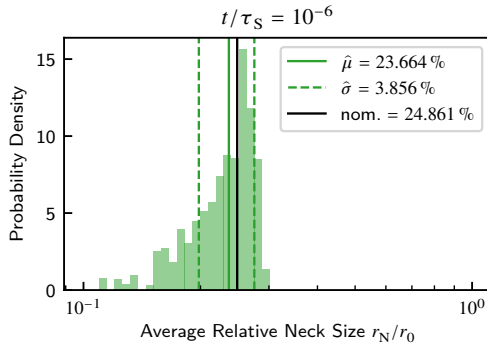
Statistical Powder Modeling

Appendix

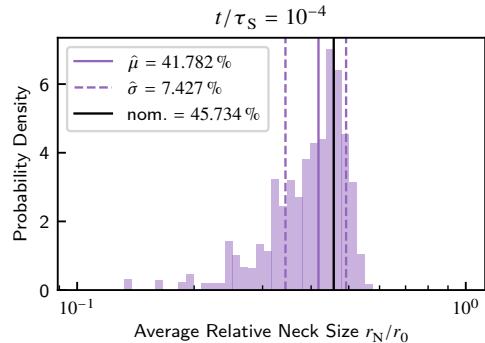
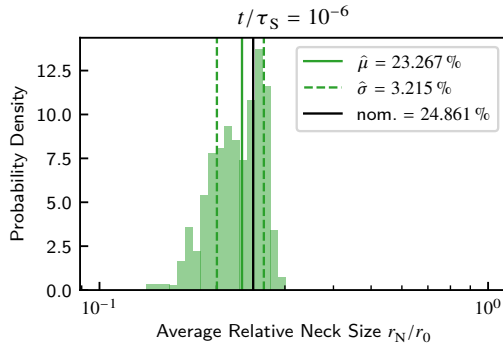
Statistical Powder Modeling

Shrinkage

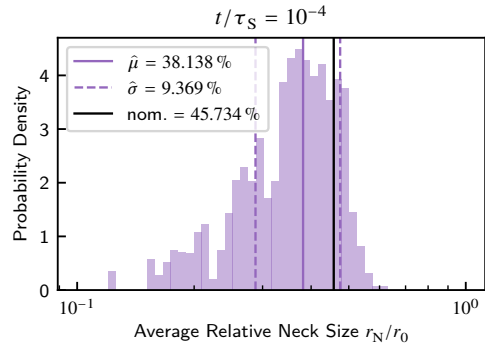
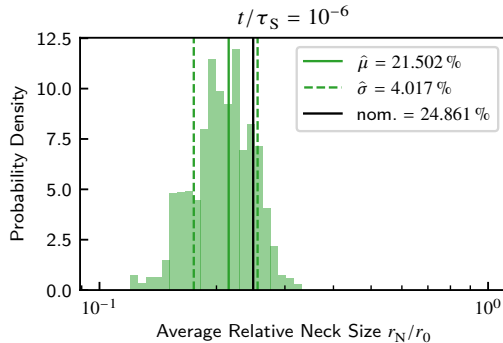
Randomized Simulation Results – Neck Size – Circular Particles



Randomized Simulation Results – Neck Size – Oval Particles



Randomized Simulation Results – Neck Size – Waved Particles



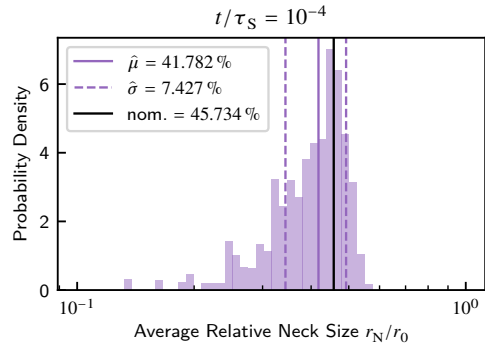
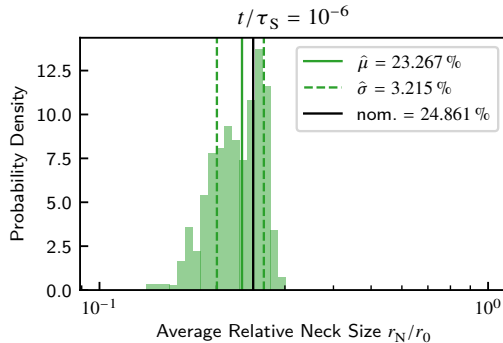
Appendix

Statistical Powder Modeling

Neck Size

Randomized Simulation Results – Neck Size – Circular Particles

Randomized Simulation Results – Neck Size – Oval Particles



Randomized Simulation Results – Neck Size – Waved Particles

